

IE5500 Dissertation

Aéropot – smart ultrasonic system for indoor plant cultivation

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aéropot



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Abstract

The world is facing an array of environmental and societal challenges such as the overpopulation, growing urbanisation, changing climate, resource deficiency and pollution to name a few. Novel solutions, that are valid, viable and sustainable, must be developed in order to deal with such issues if society, and the ecosystem as a whole, are to thrive in the future. Various new techniques of plant cultivation in a soilless environment emerged in the last few decades. Hydro/aero/ultra-ponic systems encompass many benefits from (i) substantial water conservation, (ii) all-year-round harvesting, (iii) low maintenance to (iv) cost reduction. This paper describes the research and development of a smart ultraponic system for indoor plant cultivation named Aéropot. This is a multidisciplinary project spanning across a variety of fields such as entrepreneurship, design, engineering, information technology and environmental sciences, with emphasis on practical outcomes. Although, there is an emphasis on product development, this dissertation also addresses the potential business case for Aéropot.

1. Introduction

The purpose of this paper is to present a smart ultraponic system for an indoor plant cultivation named Aéropot. It begins with a qualitative research into 21st-century's global issues of sustainable development such as global warming, habitat loss, water scarcity, eutrophication, etc. Likewise, increased levels of urbanisation force people into confined spaces with limited access to nature, which consequently creates many sociological, psychological and environmental problems in return. These were the primary source of inspiration for further research into technical solutions that can effectively and holistically tackle those challenges from a sustainability, design and business perspective. At the same time, the Aéropot solution is meant to encourage its users to find appreciation and build respect for nature, technology and underlying scientific knowledge.

The main aim of this project was to create a product that is *theoretically and practically valid, manufacturable with parts that are readily available, and unique* — meaning that it was not made before or it possesses a combination of functions and/or advantages which competitor products on the market do not bestow. The first objective was a complete analysis of the current situation on the market including an exhaustive list of all existing products in order to be as objective and transparent about the present situation as possible. The second objective was a literature review on various relevant topics such as smart technology, its potential and application, and hydro/aero/ultra-ponic systems, their function and benefits. Besides the aforementioned qualitative research, this dissertation describes a practical multidisciplinary project that crosses variety of fields such as design (e.g. the structure of the product), engineering (e.g. positioning of the individual electronic components), information technology (e.g. coding and control/monitoring elements) and a bit of chemistry. The third objective was to test the electrical conductivity of fertilizer mixed with water in different ratios to make sure control and monitoring features of the system worked correctly. The next objective was to

expand the theoretical learning and practical experience into a solid and presentable business plan. And the final and foremost objective was to develop a semi-working prototype which would serve as a learning and experimental medium to understand the technical concerns that emerged during the development process. The prototype will provide necessary feedback needed to achieve true qualities of originality, sustainability, manufacturing cost-effectivity, affordability and user-friendliness in the development phase of the final marketable version of the product.

2. Methodology

Firstly, a thorough research into changing demography, gardening market and sustainability trends was carried out in the form of a literature review. This led to the realization of the need for further exploration into other relevant topics, such as an indoor gardening, theory of biophilia or smart home technology, serving as a necessary technical base for the product development. Secondly, a detailed market research into -ponic systems was carried out to support the opportunity, leading to a sensible proposal for marketing mix. Key challenges and uncertainties were identified as well. A number of supplementary tools were utilized to identify internal as well as external strengths, weaknesses, opportunities and uncertainties (i.e. SWOT, competitor analysis, FSSD, cost analysis, etc.). Next, laboratory tests were carried out with fertilizer products available on the market and data were gathered with accessible electronic sensors. Primary quantitative data were collected during analyses in the chemistry lab. A key part of this research was the design process. The ‘Double Diamond’ framework (Appendix 1) was followed throughout the whole design process, starting with research findings or insights, conceptual proposal and finishing with the prototype of the product. Generally, the design process is specifically customized to each project, nonetheless, it is typically an iterative process where idea generation and evaluation are performed alternately until the satisfactory results are produced. Both theoretical background and practical aspects of product planning

and development were described. A selection of electronic components, CAD modelling or 3D printing were some of the areas, where the author had to build theoretical understanding and subsequently practically applied it. The product design and prototype development considered requirements of the technology and proposed design language. Since the time for conducting the experiment was fairly limited, certain theoretical aspects were omitted for the sake of practical and quick results. Design and engineering of the prototype, sensor measurements and calibration, electronics wiring, and other aspects were tested through a method of trial and error. Theoretical background of the relevant mechanisms was described in the sufficient depth. Books, journal articles, business reports and specialised websites served as secondary research sources for the literature review. Many information sources such as databanks on company websites or articles from academic institutions provide simple explanations of relevant topics whereas journal articles usually deal with very specific areas of interest without providing sufficient overview first. On one hand, many theories and practices of previous academic research were incorporated and on the other hand “less” reliable and valid internet sources helped to figure out issues in technical areas of the project. For example, many internet users showcase their experimental Arduino projects step-by-step on YouTube. Likewise, Instructables website has a wide array of do-it-yourself projects and multiple online forums host topics about -ponic systems and in-depth personal experiences of the hobbyists. The general objective was to gather as much useful information that would be applicable during the design and development stage.

3. Global issues of 21st century

The world’s human population has been steadily growing and it is estimated to reach 9.7 billion by 2050 and over two-thirds of the global population is set to live in the cities by then (United Nations, 2014, 2015; Figure 1). Additionally, more than 75% of the Earth’s land surface has been transformed as a result of pervasive human development (Ellis and Ramankutty, 2008).

“Now that humans have restructured the terrestrial biosphere for agriculture, forestry, and other uses, global patterns of species composition and abundance, primary productivity, land-surface hydrology, and the biogeochemical cycles of carbon, nitrogen, and phosphorus, have all been substantially altered” (ibid.). Population growth is also predicted to have a large effect on the demand for water which may drive up water shortages in the urban areas (McDonald et al., 2011). Besides the inevitable endeavour to provide clean water, it will be also necessary to increase the global food production by 60% to 70% in order to feed the whole population in 2050 (Foote, 2015). Furthermore, temperatures are estimated to rapidly increase in the next few decades due to the global climate change (Kaushal, Gold, and Mayer, 2017). The evaporation from reservoirs and watercourses during spring and summer is expected, meaning that prolonged periods of drought will have an impact on the agricultural productivity (Holmes, 2017). Already, around a quarter of arable land has been asserted unproductive, unfertile and unsuitable for performing regular agricultural activities (Lakhia et al., 2018). All the aforementioned environmental phenomena, and other such as sea level rise, ocean acidification pollution, etc., are interconnected, and their future interactions are complex and unpredictable (Ammar, 2017; Flörke et al., 2018). McDonald et al. (2011) advise a few solutions to decrease demands of growing human population on the environment. Specifically, innovations in agriculture and landscape management in terms of efficiency gain could save valuable water for city dwellers and land for the spread of natural habitats. Moving away from human dominion over nature’s ecosystem would mean a restoration of ecosystem structure, function and services.

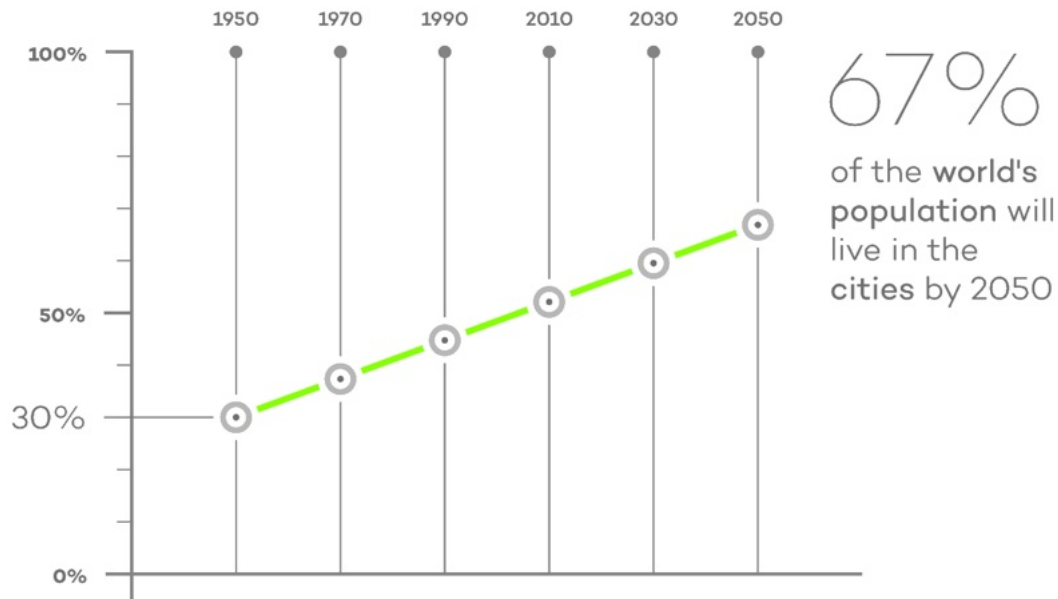


Figure 1 – Global urbanisation rate (United Nations, 2014)

On one hand, positive economic growth and industrialization are accompanied by urbanization as a demographic transition; on the other hand, it also carries some negative aspects such as polluted environment, or reduced social support (Srivastava, 2009). This makes it very likely to have adverse effects on the individual's mental health, quality of life and human well-being in general (Bratman et al., 2012; DEFRA, 2011). Nature has always been at the centre of human experience. Humans are dependent on its continuation of material production (i.e. tangible benefit) in order to survive (Keniger et al., 2013). Particularly since the post-war era, rising industrialization and urbanisation caused the movement of whole populations from rural to urban areas, allowing for more effective ways to mass-produce food and acquire wealth by exploiting natural resources. Research suggests that living (or being near) a natural setting does also have an array of psychological, cognitive, physiological, social and even spiritual beneficial effects on people's well-being (Appendix 2). The exposure and interaction with rich natural environments can evoke and enhance feelings such as subsistence, protection, affection, identity, self-enhancement, pleasure and freedom (Kingsley et al., 2009). However, the impact

of urbanisation reduced the opportunities or chances of being in contact with natural elements such as plants, animals, or even clean water or fresh air.

Residential city areas re-formed the landscape and local ecosystems into static and highly cultivated surroundings. Wildlands currently occupy less than 22% of the Earth's land surface (Ellis and Ramankutty, 2008). Less exposure to natural elements means less understanding or even interest in nature (Kaplan & Kaplan, 1989; Bratman et al., 2012). Those elements are rather replaced by modern artificial artefacts such as smartphones and high-definition TVs (Gelsthorpe, 2017). The gap between natural elements and human senses has been undoubtedly widening. The data trends suggest that rapid urbanization has confined city dwellers to acquire a more indoor lifestyle with less exposure to the outer environment (Figure 2). One of the most common ways of directly interacting with natural or green spaces in the urban setting has been gardening (Kingsley et al., 2009). As in many other areas of human activity, the attitude towards gardening has been changing, drawing us further away from the natural world.

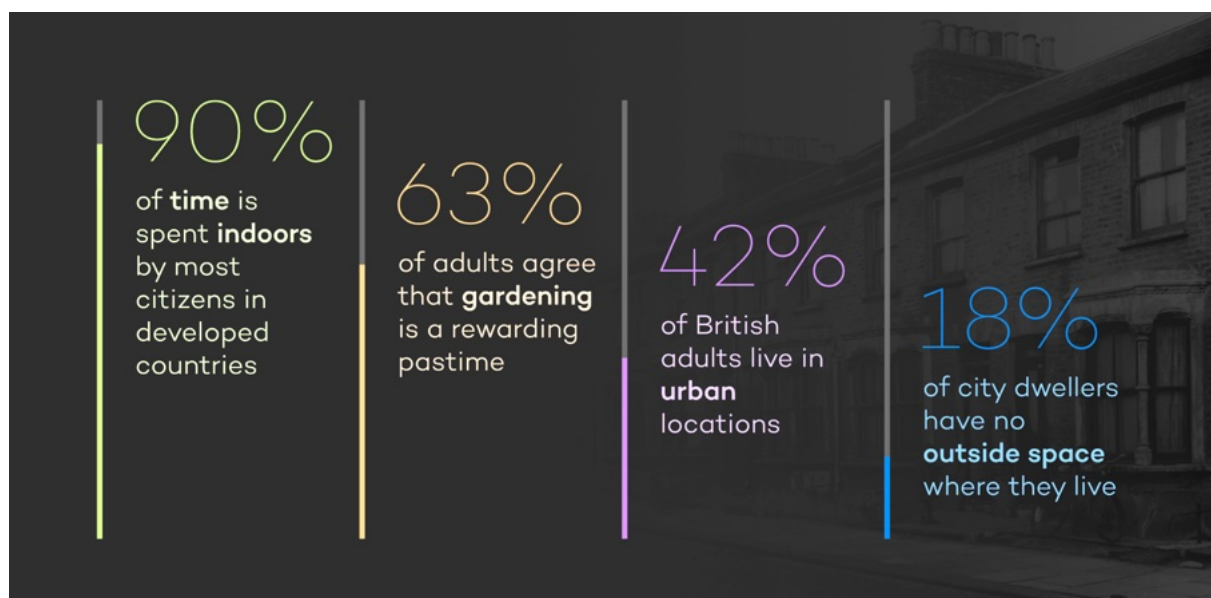


Figure 2 – Changing gardening attitudes (Mintel, 2017)

4. Overview and concept of -ponic systems

Conventional, or so-called geponic, agricultural practices (i.e. growing crops in soil, in the open air, with irrigations and active provision of nutrients) have had a broad range of impacts on the natural environment (AlShrouf, 2017). “Some of the negative impacts of conventional agriculture include high and inefficient use of water, large land requirements, high concentrations of nutrients, and especially soil degradation” (ibid.). In combination with the exponentially growing human population, and to meet its demands, production of food and urban area development will inevitably increase. Switching to more effective water-use technology in the agricultural sector could substantially reduce the urban surface-water deficit (Flörke et al., 2018). Alternative systems have been put into practice in recent years such as hydroponics (i.e. hydro agriculture), aeroponics (i.e. aerobic agriculture) and most recent ultraponics or fogponics for growing agricultural products (NASA, 2004; Jones, 2016; Folds, 2018). The soilless plant cultivation methods can be classified as opened or closed (Appendix 3). In the closed system, the nutrient solution recirculates and is adjusted according to its concentrations. The dissolved fertilizer compounds are well-adjusted in specific ratios otherwise if the balance is disrupted it could have a detrimental effect on the development of the plant. Whereas in the open system, a new nutrient mix substitutes the old one after each irrigation cycle (EI-Kazzaz, 2017). On the one hand, closed systems have the advantage in terms of their efficiency with water and nutrient usage. On the other hand, open systems are easier to control as the nutrient solution is pre-mixed beforehand and regularly exchanged, therefore minimizing the risk of “out-of-balance” scenarios.

The main difference between the conventional way of growing plants and modern methods is the utilization of nutrient-rich liquid solution rather than soil. Specifically, hydroponics is a soilless method of enclosing the root system in a dark enclosure and supplying water and mineral nutrient fertilizers: either passively (i.e. Kratky method; Figure 3) or

actively, with the use of water pump to provide an extra oxygen (Ritter et al., 2001; Figure 4). This system consists of a reservoir, nutrient pump, timer, air pump and stone, and a grow bed (Green Desert, 2014). There are different techniques growing plants hydroponically (ibid.):

- Kratky's method,
- Nutrient Film technique,
- Dripper Feed system,
- Deep Water Cultures system,
- and Ebb & Flood (Flood & Drain) systems.

The aeroponic technique is utilizing misting nozzles to spray the nutrient solution over the roots (Figure 5). It does not require any growing medium unlike hydroponics (e.g. clay pebbles, rockwool, perlite, etc.). However, the benefit of using a growing medium is the retention of air and water that allows plants to “breathe” (Nanda, 2018). Moreover, the growing medium does not necessarily have to be organic, simple polymer foam can serve as an anchor and an effective absorbent of moisture and oxygen.

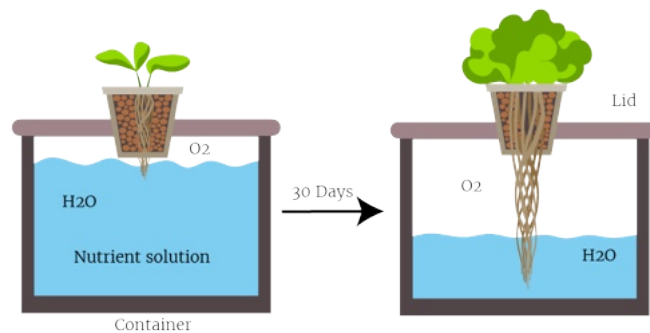


Figure 3 – The Kratky method (passive) hydroponic system (Green and vibrant, 2018)

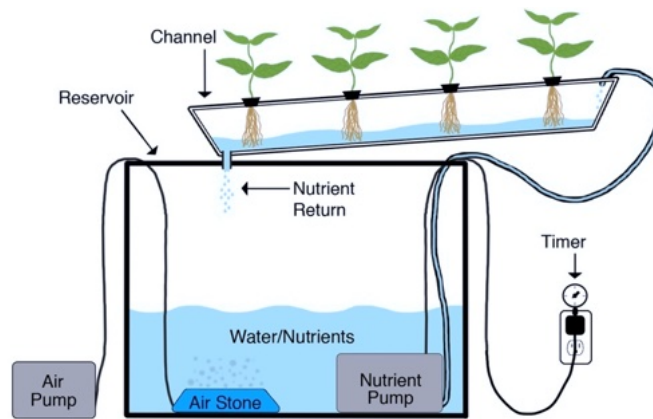


Figure 4 – The active hydroponic system (Green Desert, 2014)

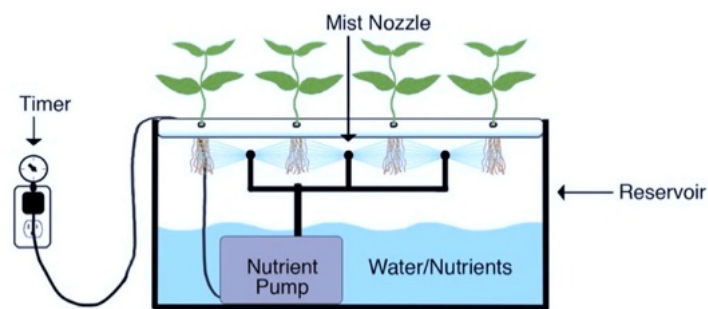


Figure 5 – The aeroponic misting system (Green Desert, 2014)

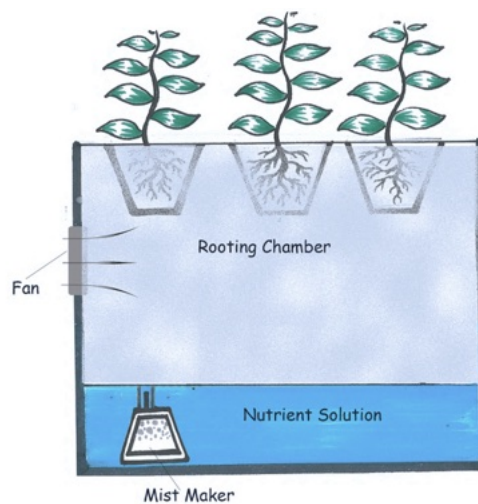


Figure 6 – The ultra-ponic misting system (Hydroponic Passion, 2016)

In theory, there are three types of aeroponic systems according to the size of water droplets (Shubham et al., 2018):

1. Low pressure Aeroponic Systems (soakaponics)
2. High pressure Aeroponic Systems (true aeroponic systems)

3. Ultrasonic foggers

High-pressure atomization nozzles deliver nutrients through a mist to the roots between 10-100 micrometers (Lakhia et al., 2018). These nozzles are driven by a pressure air pump with PSI of 80 to 150. Soakaponics (or low pressure aeroponics framework) instead utilizes a submersible water pump that flushes water through standard sprinklers. Ultraonics or fogponics (also called ‘dry-fog aeroponics’ by some) is the latest method of growing plants in a soilless environment (Figure 6; Hikosaka, 2016). It is very similar to aeroponics (sometimes considered as a subtype of aeroponics), with the difference of using ultrasonic mister or atomizer instead of a nozzle (Hydroponics Systems, 2012). In principle, the ceramic disk inside the mister oscillates at high frequencies and breaks the water into very small pieces that float in the air (i.e. the mist). Within the aeroponic/ultraonics systems, the differentiation comes down to the technique or technology of irrigation and nutrient intake. Essentially, both methods encompass the process of atomization — breaking or reduction of liquid molecules into fine droplets (Lakhia et al., 2018). The size of the droplets and the speed at which the droplets are released onto the surface of the roots affect the development of the plant. “The root's mist collection efficiency depends on its filament size, drop size, and velocity” (Clawson et al., 2000). Hikosaka (2016) conducted an experiment with leaf lettuce plant and found out that optimally controlled flow rate of the nutrient mist can “...accelerate the root growth adaptation, thereby leading to improved growth in the early stages after planting.”

All aforementioned -ponic systems have many considerable advantages over conventional plant cultivation methods (AlShrouf, 2017):

- the soil is not necessary,
- higher nutrient and water use efficiency due to control over nutrient levels,
- no nutrition pollution is released into the environment,

- supply of fruits and vegetables continually throughout the year without dependency on the outside conditions,
- an ability to practice the vertical farming production which requires less water and space,
- and they are all considered as safe and environmentally friendly methods for plant cultivation.

Furthermore, hydroponics uses about 13 times less water, and a fraction of space, needed to produce an equivalent yield amount in comparison to traditional agriculture (Figure 7). Nevertheless, aeroponics can reduce the water as well the fertilizer and pesticide usage even further and at the same time increase the yield per plant around 70% when compared to the total production in hydroponics (Ritter et al., 2001). Some tests claim that aeroponic systems can save water up to 99%, nutrients 50% and 45% less time than with soil-based cultivation methods (Lakhiar, 2018). Aeroponically grown plants can uptake more vitamins and minerals, therefore making the plants potentially more nutritious and healthier (AlShrouf, 2017). In a search for water, nutrients and air, the majority of the plant's energy is spent on the development of its roots. In close field cultivation methods, the availability and concentration of these elements can be well balanced, hence providing the plant exactly what it needs (Nanda, 2018). At the same time, -ponic systems avoid enormous risks and uncertainties from biotic and abiotic stresses, such as pest attacks, droughts, floods and high winds that are so prevalent with standard agricultural practices (Lakhiar et al., 2018). The water loss due to evaporation is much less in hydroponic systems, so it can also work effectively in drought-stricken areas. From a practical point of view, the system can be installed in various places such as greenhouses, balconies, roofs of buildings or even in lands unsuitable for cultivation with extremely cold/hot climate (i.e. deserts or tundra; El-Kazzaz, 2017).

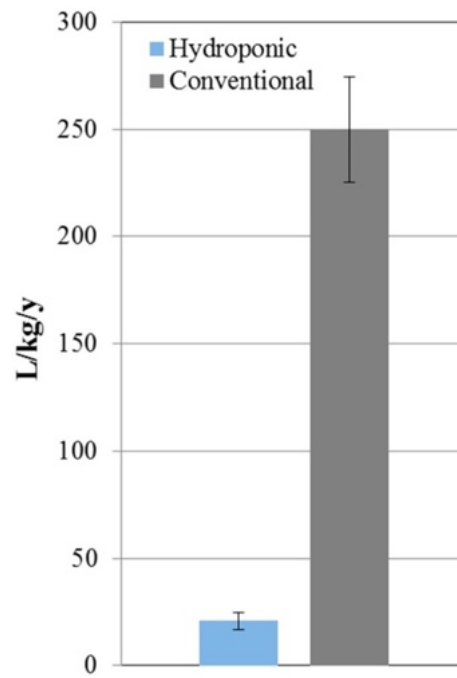


Figure 7 – Annual water use of lettuce using hydroponic vs. conventional methods (Barbosa et al., 2015)

Aeroponics specifically offer multiple advantages over other methods (Hayden, 2006; Chiipanthenga et al, 2012; Giurgiu et al., 2015; AlShrouf, 2017). Firstly, the increased aeration of nutrient solution delivers more oxygen to plant roots and thus stimulates the growth and reduces rooting times. The root chamber offers a large space for a free extension of the roots, whereas in the soil cultivation method, the root system is constantly obstructed with fragments. Aeroponics uses less water and nutrients because the plant roots are sprayed at intervals using a precise droplet size that is utilized efficiently by osmosis (Lakhia et al., 2018). Also, as the fluid solution is actively sprayed or misted onto the roots, there is less chance of a pathogen formation. The seedlings of the plants do not have to grow sprouts before they are placed into the aeroponic container since there is a continuous and uniform influx of moisture for them to germinate. This cuts down steps in the cultivation process, saves time and attention of the user. Lastly, it permits multiple harvests from a single crop.

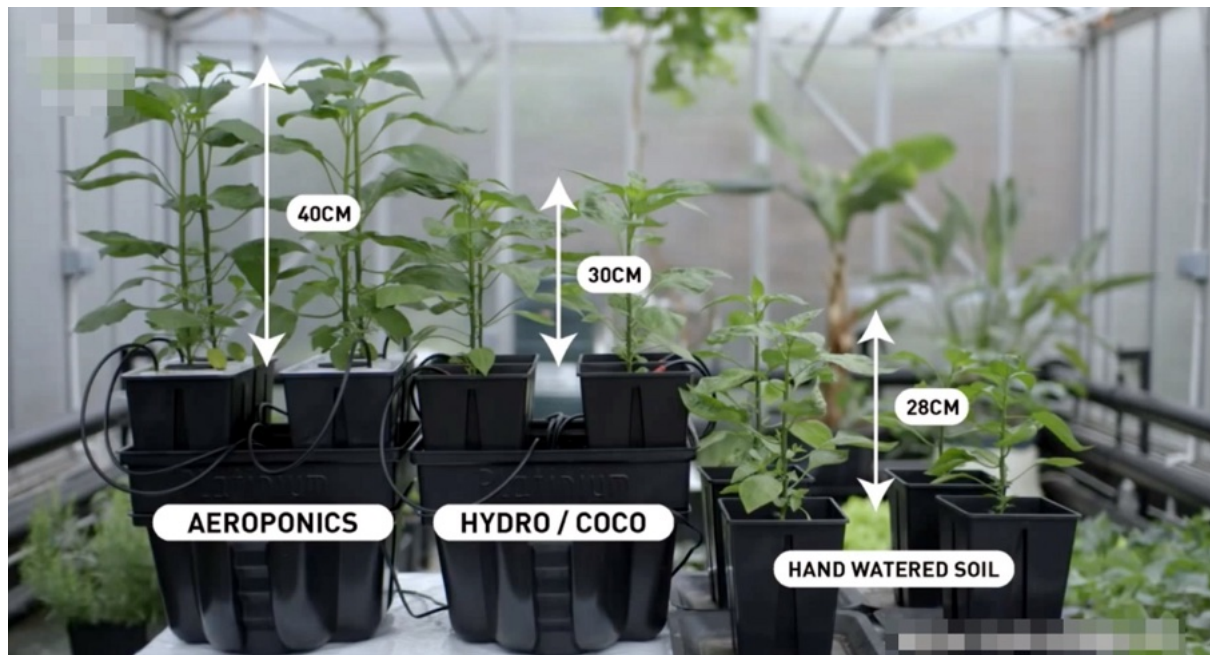


Figure 8 – Comparison of different plant cultivation methods (Hydroshow TV, 2014)

5. Ultraponics challenges and assessment

There is an ongoing discussion and not much academic research available in regard to the use of ultrasonic misters for soilless plant cultivation. Ultraponics uses even less water than hydroponics, it has higher yields on average and it is very versatile to set up. The development of the root hairs as well as the fog trapping efficiency is higher (Hikosaka, 2016). Maintenance of the misting units is much easier in comparison to nozzle heads. Nevertheless, there is some downside to this technology. The ultrasonic mister heats up the nutrient solution, thus affecting the nutrients and could potentially damage the roots. Another issue with using foggers is that the plates tend to clog with mineral build up (Lakhier et al., 2018). Most importantly, many academics and hobbyists argue that ultrasonic misters actually only produce water droplets under 5 μm (microns) of size (Electro Tech, 2013; ResearchGate, 2015). More concrete evidence comes from Phanphanit (2011), who tested regular off-the-shelve atomizer working at 1.72 MHz. The results show that it produces droplet sizes in the range of 6-12 μm . Droplet sizes in the range from 5 to 15 μm may not be too ideal as larger or heavier nutrients may not

get carried to the roots (Clawson et al., 2000). It has been suggested that the optimal value of the droplets should be around 30 μm (ibid.). A quick practical observation test was carried out with the purchased ultrasonic mister.



Figure 9 – Large water droplets (left), microscopic droplets or the mist (middle), and condensed water (right)

First, the mister was inserted into a stainless-steel bowl filled with enough water for it to work correctly. Then, the ultrasonic atomizer was turned on for a quick session of misting (less than 60 seconds). A series of photos were taken (Figure 9), where the variation of droplet size is clearly displayed; from a levitating mist to rain-like drops and condensed droplets on the surface of the bowl. As mentioned, the main concern is that either the droplets are too small to carry any nutrient with them or they are too large to attach onto the roots. “The larger droplets permit the less supply of the oxygen availability in the root zone. The smaller droplets produce too much root hair without developing a lateral root system for sustainable growth” (Lakhier et al., 2018). Although more accurate testing is necessary, a simple observation showed that the ultrasonic atomizer generates droplets of varied sizes; Figure 10 illustrates this notion well. Hence, it is almost certain that the water and nutrients are captured by the plant roots when using an ultrasonic mister.

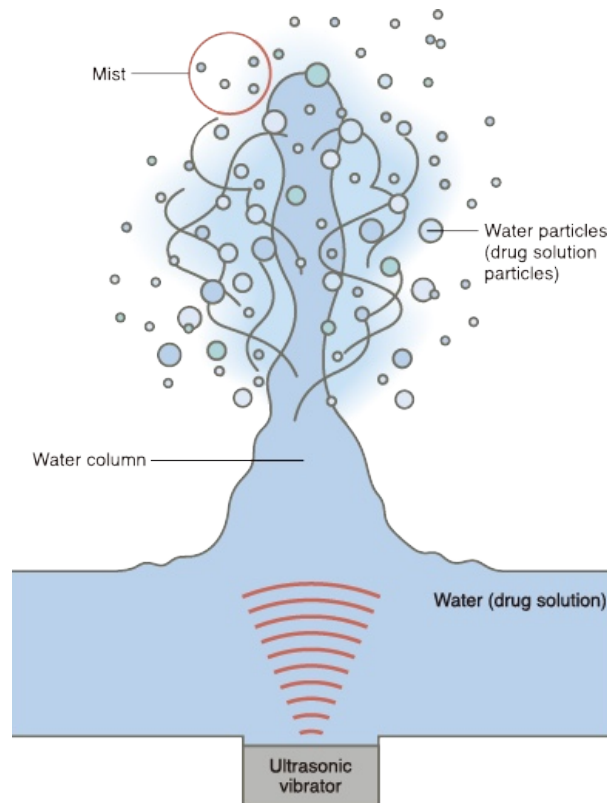


Figure 10 – The principle of ultrasonic atomization (Honda Electronics Co., 2016)

Adjusting the vibration frequency can lead to reaching the desired size of droplets; lower the frequency, the larger the size (Phanphanit, 2011). Potentially, a misting nozzle could be used instead as it can create a mist with droplet sizes changing according to the diameter of the outlet orifice/pin which is more consistent in terms of volume and droplet size as well as much easier to control. On the other hand, it requires an adequately powerful pump and therefore substantially increases power consumption levels. Nevertheless, it is a good consideration for the future design.

6. Market trends

The market is ripe for innovative garden and home solutions (Euromonitor International, 2018a/b; Mintel, 2018; Appendix 4). It is expected that companies will invest in products that allow users to save time and money, and curb the effort needed to take care of the garden. They are re-thinking their product offerings in favor of indoor gardening. This will include a focus

on Internet of Things (IoT) and automation of technology. Gardening solutions such as hydroponics and “smart gardening” will drive this innovation. Likewise, the trends toward green and sustainable cities and healthy lifestyle among their citizens are creating new opportunities in the market. Increasing the consumer focus on healthy eating habits spurs demand of gardening products with features that embrace such lifestyle.

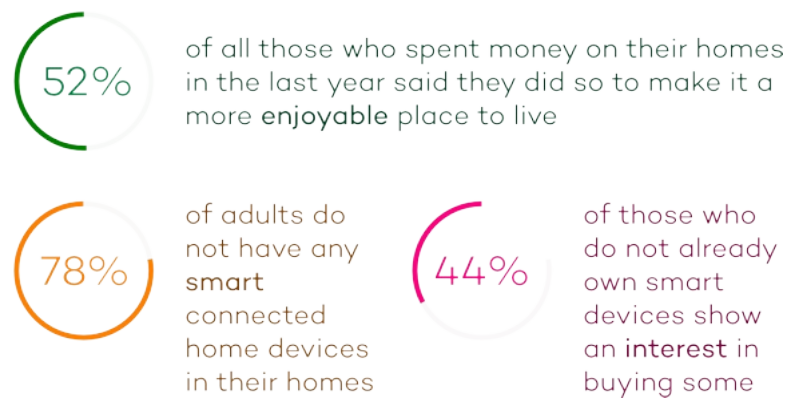


Figure 11 – Consumer trends related to spending habits for the home (Mintel, 2018)

This paragraph provides justification for the selection of main target groups. As the garden is increasingly seen as a place for socializing and relaxation, plants are utilized mainly for a decoration only (Garden Media Group, 2017). Consequently, the cultivation of plants is moving indoor, in order to reconnect with nature and counter the pollution. Furthermore, owning a private garden is becoming more and more of an exception. Specifically, with the Millennial cohort who are moving to live in the city apartments. They are tech-savvy and looking for creative ways to cultivate organic plants. According to the Garden Media Group (2017), 37% of Millennials are already growing herbs indoors, that is 9% more than Baby Boomers, who are presumed to be more active in outdoor gardening. Majority of parents also think that their children should be involved in activities centred around nature, gardening and healthy food. “There is significant educational value in helping children understand how food is grown, as well as the nutritional benefits of eating fresh salads, making families a key target for urban and windowsill planters” (Mintel, 2017). Young children have a fascination with the

immediate, and an undying curiosity that requires direct sensory experience (Sobel, 1996). So, an object with unordinary properties that combine technology and natural elements can interest them far enough to induce attitudes toward appreciation, nature conservation and further learning about relevant topics. Another context, where natural features may improve a variety of conditions, is an office space (Smith et al., 2011). Indoor office plants, in particular, create healthier and happier workers, increase productivity, lower healthcare costs and absenteeism (Garden Media Group, 2017).

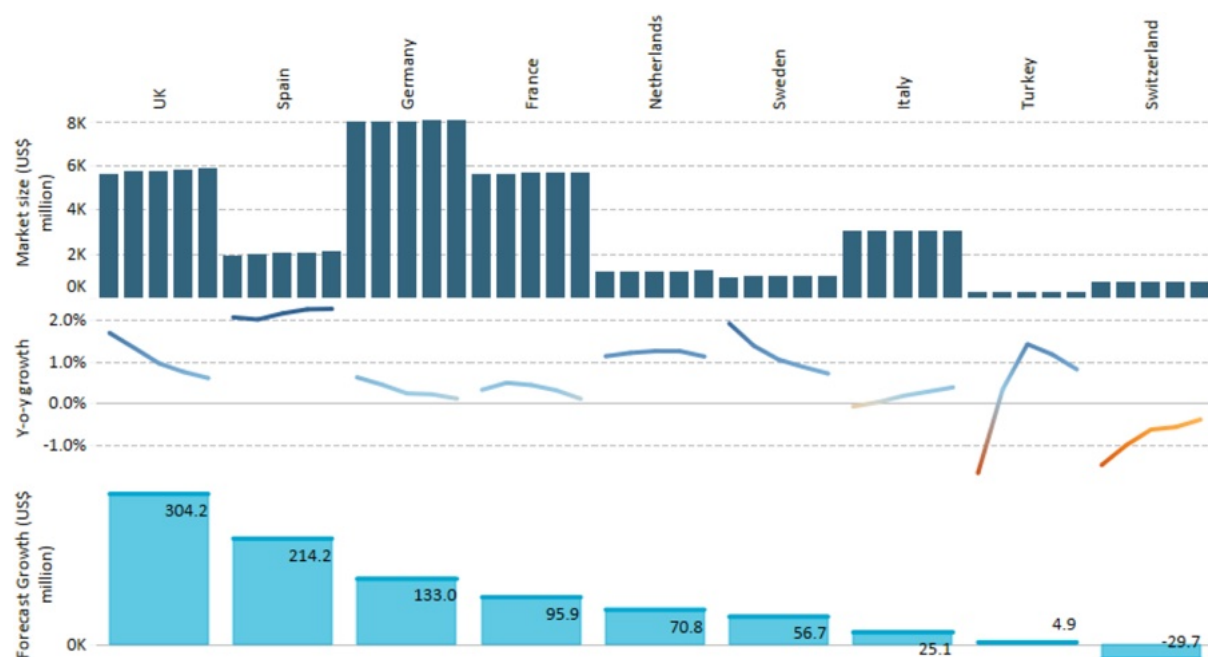


Figure 12 – Gardening market size and growth forecast in Western Europe 2016-2021 (Euromonitor International, 2018a)

North America is the leader in the global gardening market and is set to steadily grow. Germany and the United Kingdom are the largest gardening markets in Western Europe, accounting for a 37% value share (Figure 12). The UK has the most lucrative market; consumer spending is forecasted to grow by 22% and reach £7 billion by 2021. There is also a great potential in regions such as the Middle East and Africa along with the Asia Pacific which are the fastest-growing markets. Japan and China have had historically a deep and long relationship towards gardening and art, and they are on the rise again (Sun, 2016). There is also a great demand for pollution protection products such as masks (83%) and air purifiers (56%) in China (Mintel,

2016). This indicates that protection against pollution is a major concern, and consumers generally respond by using houseplants (52%) and air quality index monitoring (33%) in households too.

Within the consumer product market, there is a tendency towards universal and simplistic visual design. Electronic devices are primarily designed in such ways in order to use as least materials and components as possible and avoid unnecessary activities during the manufacturing phase. At the same time, the emphasis is on having simple physical form or casing in order for the product to seamlessly fit into any indoor environment. Control and monitoring elements of the device are [or should be] positioned and offer functions that the user understands without lengthy learning process. Some examples of effective product design come from design school of Dieter Rams or more recent portfolio of smart devices from Apple company, where the design is prioritized among other aspects of the product (Mokhov, 2011). Another movement is represented by a company like IKEA for example. It is the forerunner for the democratisation of design, producing products that are beautiful, accessible, and affordable and upholding design principles that use natural materials, clean lines, and straightforward construction.

7. Concept development

The conceptual proposal of Aéroport system stems from the prior research, careful observation of the existing market, and author's judgement. Much of the work during the design process consisted improving difficulties around the practical use of the product, but also understanding of the implicit needs of the user, asking questions such as (Boeijen et al., 2016):

- What would the user want?
- How would the user use it?
- How would/should the user feel while using it?
- What emotions to evoke?

In the process of answering all those questions, drafts and other notes were generated to develop the final idea (See Appendix 5). Once the purpose and function of the product were established a bit further, the form of the physical object and experience of the interaction system were explored. A large moodboard of desirable shapes and settings was created (See Appendix 6) and the two most prominent and available apps on the market (in the field of -ponic planters) were analysed (See Appendix7). Figure 13 shows the final design concept, and further details regarding the R&D are presented in Appendix 8.

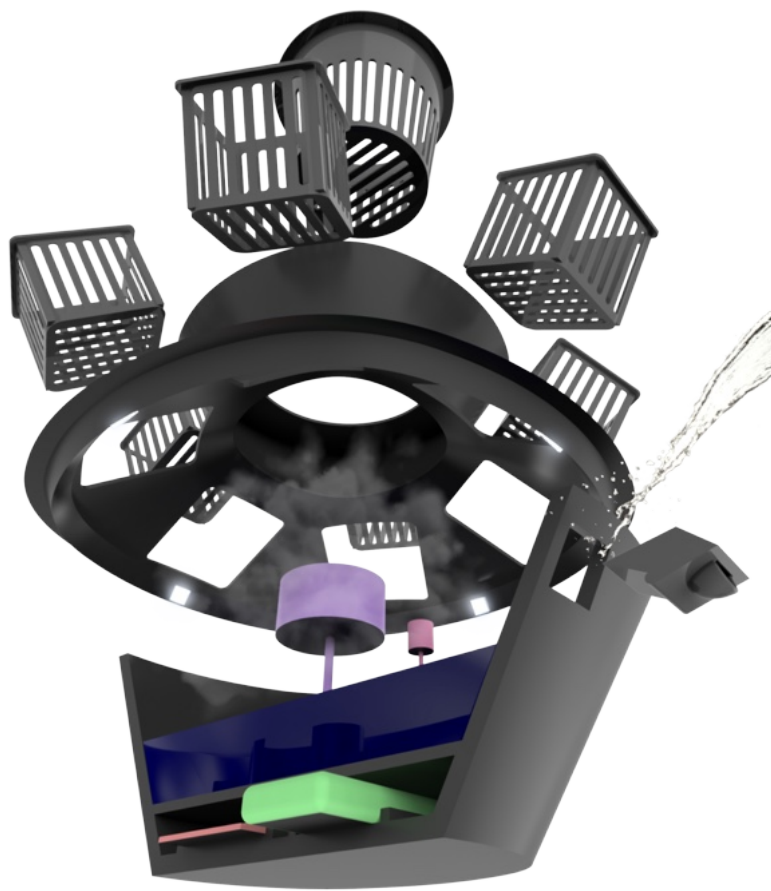


Figure 13 – Full render of the Aéropot device concept

In summary, Aéropot is a smart aeroponic device that fits into any environment thanks to its universal and elegant design. It aims to provide full, closely-monitored control over the plant's vitality. At the same time, it serves as a tool for ecological education. Combination of high functionality and clean natural form induces emotionally durable bond between the user

and product. It encourages people to find appreciation and build respect for nature, technology and underlying scientific knowledge. Aéropot has four main functions (Figure 14):

1. Flower caring/cultivation; adding aesthetic/ornamental value
2. Air purification; oxygen production
3. Edible plants (e.g. vegetables, fruits, and herbs)
4. Medicinal plants

“Although it is a common practice nowadays to grow the numerous commercial and specialty crops hydroponically, some species will grow better than others. For instance, hydroponics can be ideal for the cultivation of leafy vegetables such as lettuce and herbs, fruit vegetables like tomato and cucumber, and some ornamental plants” (Ammar, 2017).



Figure 14 – Ways to use: growing plants for aesthetics, air purification, edibility and/or recreational purposes

Aeroponic systems and growing plants requires a lot of energy, time and resources in general. Aéropot allows for maintaining perfect levels of nutrients in the spray which is the most critical criteria for the aeroponic system to be successful. Basic sensors are integrated into the chassis of the device, where optional modular sensors can be added anytime, which allows for detailed system monitoring and prevention. At the same time, classical aeroponic systems are not made to be an object of aesthetically pleasing form, rather they are made for industrial,

higher-yield purposes. The conceptual model of the planter was 3D printed to see how the object looks (and feels) in real-life (Appendix 9). The product design of the planter was intended to accommodate all the necessary functional features (e.g. pots, lights, etc.) but also, to have a visually pleasant outlook. Furthermore, the few aeroponic systems, specifically for households, are not very energy-efficient since they use water pumps, whereas Aérospot is using inexpensive but very effective ultrasonic misting technology. Lighting, misting, measuring—all of it—is controlled by an internal logic board linked to the smartphone app (Appendix 10). The app is continuously sourcing the data from the planter through cloud services. The first time the customer uses the Aérospot, the app must be pre-configured with the name of plants that are planted into the pot and set the positions that correspond to the ones on the device. This allows for precise control and delivery of the notification about the right plant. The next step in the initial configuration of the Aérospot device is the mixing of the nutrient liquid solution. The app calculates the best composition of the fertilizer according to the input data and ‘best-practice’ scenarios. The solution is poured into the device and pots are filled with clay pellets and seeds. Once the pre-configured program is started, the device will take care of itself. It mists the roots according to the chosen mode (i.e. eco, moderate and rapid). Furthermore, it notifies the user if there is a requirement for maintenance or a need for another intervention. The user is frequently queried regarding the state of the plants and an explanation of what is happening in each plant phase is presented. As well as other information about the underlying theories (e.g. ecological cycles or even how to deal with global issues of the 21st century as an individual) and practices (e.g. what is aeroponics or what necessary nutrient substances are required and why).

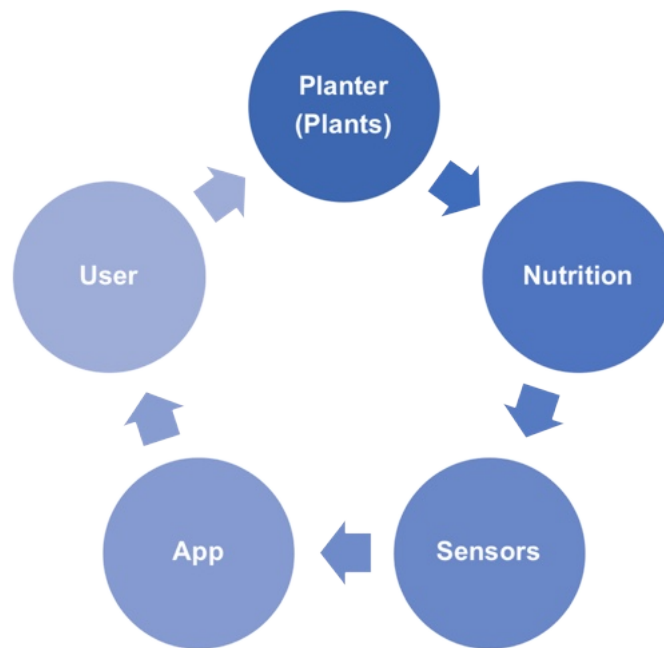


Figure 15 – Aéropot system

Aéropot is not only a planter and an app, it is a comprehensive system of products and services (Figure 15). It is meant to nurture interaction between the plant and the user through the use of technology. It starts with the planter and the plants which require nutrition and optimal environment, consequently, the environmental factors are monitored by the sensors passing the information onto the user, who can further act on the given information. “Ultimately the roots get plenty of moisture, fresh oxygen, and nutrients in the growing chamber. A well designed [aeroponic] system provides a good balance of all three at the same time” (Sethi, 2017; Figure 16). Other parameters are important to be constantly monitored such as temperature, humidity, CO₂ levels, light intensity, ventilation, pH and other (EI-Kazzaz, 2017). And it is this chain reaction starting with the environmental changes, registered by the sensors, pushed notification to the app and all the way to the user. The ultimate aim of such considerations is to guide the user in a simple and intuitive way to maximise yield and quality of the cultivated plants.

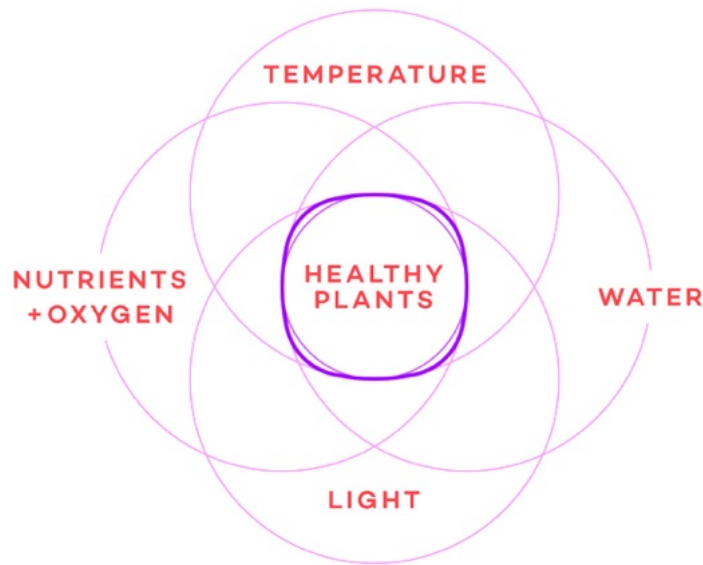


Figure 16 – Healthy plants are a combination of multiple factors

The full FSSD analysis was carried out (Appendix 11), with consideration of circular economy (Appendix 12) as a ‘checklist’ for the environmental and social sustainability factor. Most of the sustainability principles were not violated. On the contrary, the Aéropot solution would positively contribute to the environmental challenges. However, only a small-scale. By utilizing resources more efficiently and separately from the natural biotopes (e.g. fewer fields cultivated), we can decrease the pressure put on the natural system by removing or substituting the damaging methods of resource extraction. Anyhow, the material sourcing and manufacturing processes are likely to have the largest impact when the whole product life cycle is considered. In particular, the plastic casing and electronics inside. It is almost impossible to produce electronic circuits and other parts without any environmental impact. Nevertheless, the synthetic polymer materials (PP/PE) that were chosen for the casing of the product could be replaced with another material. Bamboo, for example, is a natural material with the right properties. It is cost effective, aesthetically pleasing and already used in many other gardening products (Figure 17). Moreover, it increases the sustainability angle of the product and appeal for the customer.



Figure 17 – Standard planters made from natural bamboo and partially recycled bamboo (Iron & Clay, 2018; Hema, 2018)

8. Design process

The conceptual proposal supported the efforts of understanding underlying phenomena, mechanisms and design assumptions. As insights were gained through conceptualisation, appropriate refinements and changes were possible could be done. At this point, everything was still only theoretical, so a proof of concept prototype was further developed.

8.1. Planning

The whole process of developing the prototype required considerations from various new angles in order to have a functional unit in the end. That includes engineering (e.g. electronics, wiring, durability, etc.), design (e.g. product, structure, positioning, communication, etc.), IT (e.g. code, protocols, connectivity, etc.) and even a bit of chemistry (e.g. fertilizers, TDS/pH levels, etc.). The first and foremost task was to identify the limitations of the study/project. That includes: the limits of time to conduct the research and tests; resources, facilities or tools that were available to the author (e.g. labs, workshops, academics, industry experts); the constraints of the technology (e.g. maximum build size of the 3d printer); availability and pricing of the electronic components.

The author was able to use workshops at Brunel University, including wood, metal and 3D printing workshop as well as design computer labs and chemistry labs. With the help of academics and technicians, hands-on tasks and necessary tests were carried out. The process

continued with a search for available technology, consideration of an ideal combination of parts and positioning of the components in order to bring maximum effectiveness, intuitive product design, and last but not least smooth user experience. Another important consideration was the size; the final prototype version of the planter was meant to be compact in order to be portable and light. The structure of the planter should effectively leave no inner space empty unless compromising the product functioning or negatively affecting the plant growth.

8.2. Electronic components

One of the crucial pillars of this project was to build a prototype with available and functional electronics. So, most electronics were searched and sourced from an online Chinese marketplace called Aliexpress as it provides with an extensive array of low-priced products. First, different types of LED strips were purchased for testing (e.g. blue/red diodes, full spectrum and warm white light) and blue/red LED halo rings (Figure 18). All claimed to be waterproof; some certified IP65/67 and some had no certification.



Figure 18 – LED strip (left), halo circle (middle), and chip lighting (Aliexpress, 2018)

Hydroponics and cultivation of plants is a very extensive topic with many approaches, different technologies and expert opinions. This is especially true when it comes to the provision of light to the plants. In general, the balance of the RGB light channels should lean toward red and green with a small percentage of blue (Figure 19). A proper ratio of different light colours is important for healthy growth of the plant. LED lighting is also preferred over

the classic incandescent or fluorescent (CFL) bulbs as they are much more energy efficient, radiate less heat and have longer lifetime (MIGRO, 2017).

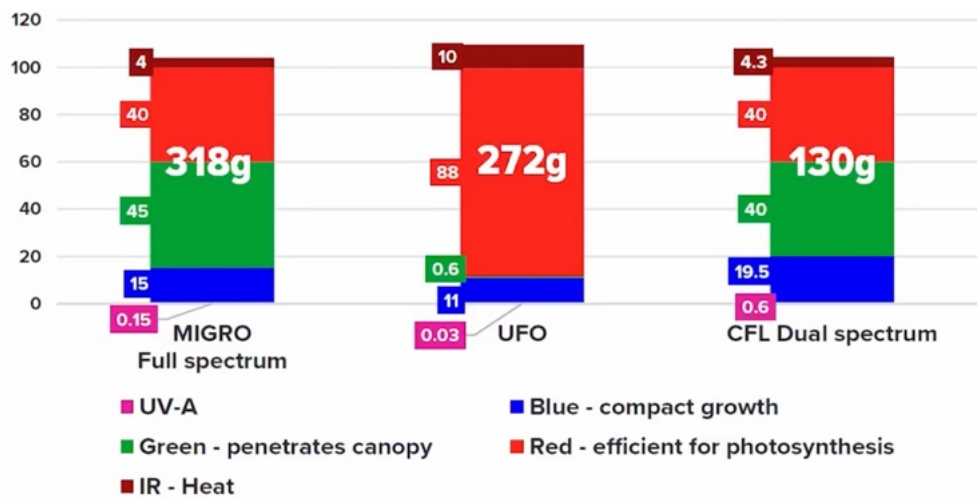


Figure 19 – Full spectrum grow light comparison; yield weight (MIGRO, 2017)

As it was mentioned in the previous chapter, plants require a very specific set of environmental and functional properties to grow effectively. Light is usually provided by the sunlight during the day, however, artificial lighting can complement and improve the growth of the plant. Light provides energy for the plant's growth and can also increase the surrounding temperature if necessary. Next, a fan was used to ventilate the inner chamber in order to deliver oxygen to the roots. The last requirement for a vigorous plant growth were the nutrients. Moreover, a research paper by Pala et al. (2014), who proposed a complete open-source autonomous greenhouse system, list an array of all vital data points:

- root temperature
- light intensity
- humidity
- oxygen availability
- nutrient intensity/composition = TDS/pH
- atomization frequency, spray time and interval time

To sustain optimal levels or concentrations, electronic sensors were chosen that are specifically constructed to detect and measure most of the aforementioned physical environment conditions. Figure 20 shows all the commercial sensors that were selected to be eventually incorporated into the structural design of the Aéropot prototype.

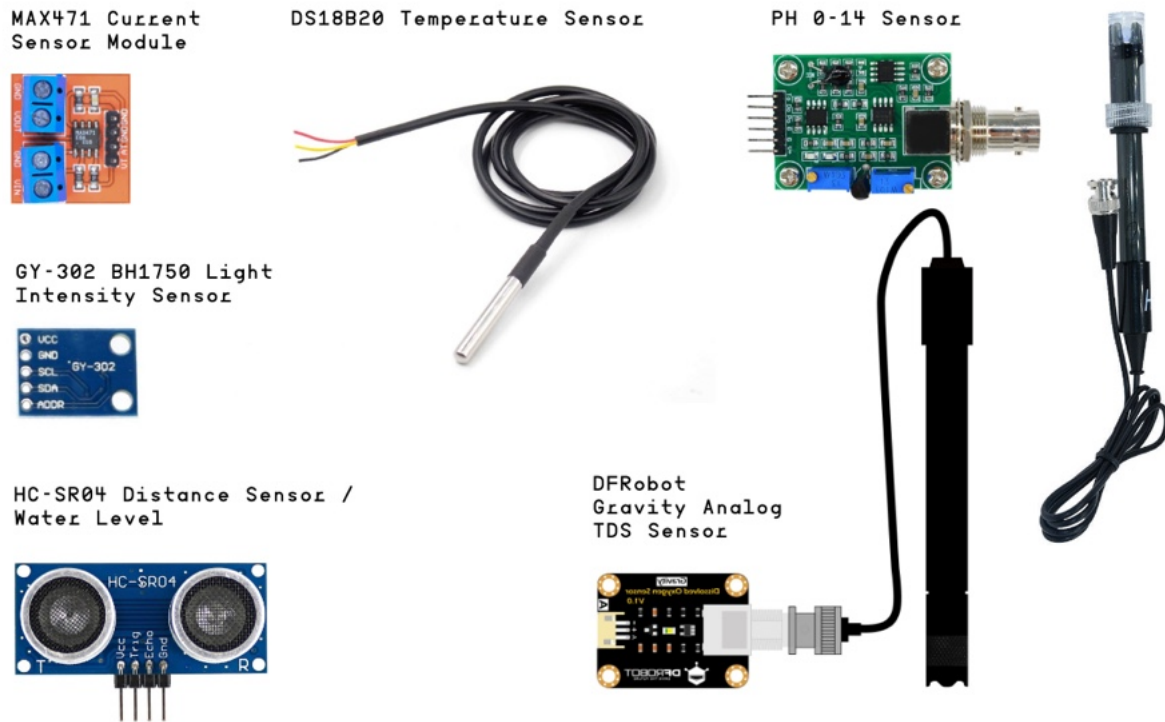


Figure 20 – Set of commercially available sensors for Aéropot prototype

8.3. CAD modelling

Once the electronic components were gathered, it was evident that the conceptual design proposal would not go well with the dimension and structure of the electronics. So, a simpler, more adjustable design, had to be proposed. The visual outlook of the planter was designed in order to be ergonomic and visually pleasing for the user. The author used various sources of inspiration during the design process. The curves and ‘outer’ feel were inspired by the design of modern consumer electronics, but also by the external outlook of cars, furniture or product concepts. Tools in the gardening sector such as pots or trays as well as existing and conceptual -ponic systems were considered from the visual standpoint. The design analysis of products in

the same market segment that Aéropot is positioned in (i.e. competitor analysis) also guaranteed that it would have the highest standing in terms of visual outlook. This is a very important differentiation aspect of Aéropot. Online platforms such as Pinterest, Behance or search engine Google Images were used to gather Figures of product design features (Appendix 13). Eventually, the aggregated database of images was sorted and analysed and served as an inspiration moodboard for the planter's design language. Further modifications had to be done to satisfy and balance it with the engineering side of the product. For example, the outer casing's final outlook was very dependent on the size and positioning of the electronics inside as well printing capabilities of the 3D printer. All the design considerations were firstly sketched on paper, evaluated and consequently modelled into a solid using a computer-aided design software (CAD) called Solidworks from Dassault Systèmes (2018). Keyshot from Luxion (2018) is a 3D rendering and animation software that was used to create photorealistic product imagery for persuasive visual presentation (Figure 19; Appendix 14).



Figure 21 – Aéropot planter prototype design

The actual physical outlook and construction of the final model deviated from the proposed concept. The model had to be simplified, not only due to the limiting size factor of the 3D printer, but also because of the arrangement of the components inside. It was necessary to leave a certain space for “errors” as the prototype was meant to be used for testing a variety

of different electronics and mechanisms. So, in practice, there would be spaces with larger dimensions than are required for fitting the electronics. The planter is meant for a very practical use, taking into account modularity aspects so components can be easily accessed and substituted. Besides taking into consideration the technical limitations and practical outcome aspect, there was a strict emphasis on aesthetics, intuitiveness and user experience (Appendix 15 and 16). Firstly, some inspiration for the external outlook and for the arrangement of internal parts was simply acquired through observation of existing products or product concepts. Secondly, the concept of golden ratio was applied during the Aéroport's design process. It is known as one of the oldest concepts in aesthetics in which people prefer a certain proportion (Levin, 2015). The golden ratio and related Fibonacci number patterns are frequently occurring in nature (e.g. seashells, flowers or human body) — creating pleasing, natural looking compositions. Every component of Aéroport planter was designed with the intention to follow a certain mathematical proportion (Appendix 17). However, some ratios had to be compromised for the prototype due to the technical limitations. A concept of continuous curvature variability (i.e. C2 continuity) was incorporated into the product's design language to produce smooth surfaces and edges (Fong, 2016). Moreover, the main shape of the planter takes a shape of a 'squircle' which is an intermediate of a square and a circle. And lastly, the casing should have subtle shades of colour and soft matt surfaces.

8.4. Wiring and coding

The wiring schematic was constructed with technical consideration of each component and its ultimate purpose (Appendix 18 and 19). Classic Arduino Uno R3 microcontroller board is the central unit for communication between all the components; it is a plug-and-play tool for rapid system prototyping and experimentation (Arduino, 2018). The Arduino is using a set of C/C++ functions that are then compiled directly into the board's internal memory. It is surrounded by a large number of accessible resources, like example code or step-by-step DIY projects,

provided by the growing online community. Therefore, it was fairly quick to achieve the desired functions. For each programmable component – mostly sensors – example code was found online in order to quickly test it and understand its functionality. Other programmable logic was not too complicated to construct.

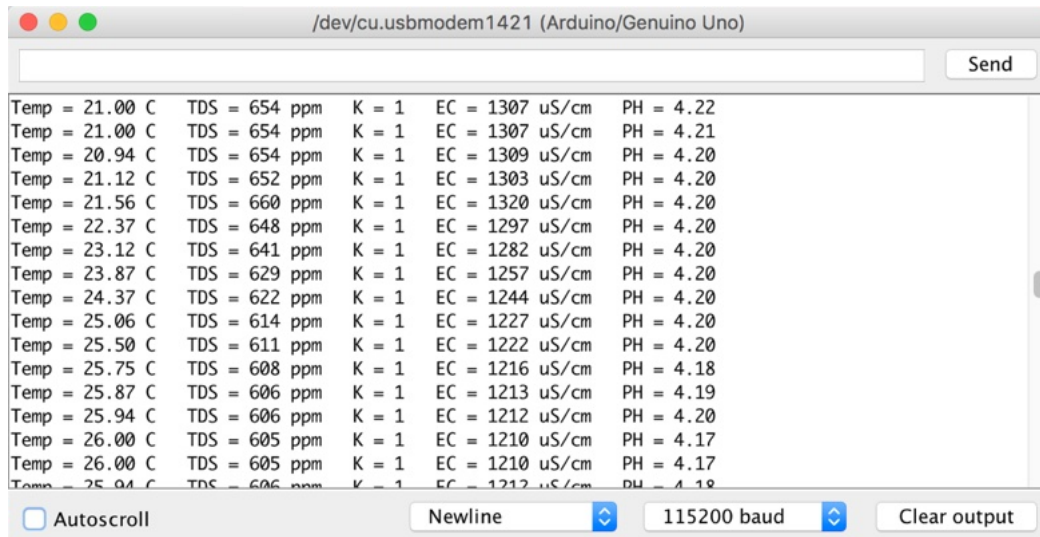


Figure 22 – Example of sensor readings in the Arduino IDE's serial monitor

The lights, mister or fan used simple commands to trigger desired output voltage on an analog pin and further regulated through a transistor. The final code controls and monitors the peripherals (i.e. output components) and receives information about the state of the environment through sensors (Figure 22). Specifically, there is a set of logical conditions or protocols formed within the code that process input/output information (Appendix 20). For example, when the levels of the TDS go below or over the established threshold, the user is notified and advised how much nutritional mix to add; either specific NPK ratio or volume of fertilizer mix. This happens in real-time. The obtained data from the sensor modules are utilized to optimize controllable properties of the environment such as temperature, ventilation, lighting and nutrient solution intake. To summarize, the main purpose of the final algorithm is to adjust the peripheral output performance in accordance with the [sensor] input data and inform the user. Moreover, the user has an option to control the variables manually. There would also be

considerations for when the product is not in use, the app would recognize the inactivity and ask the user what the reason for the inactivity is. In a case of malfunction, the user could order a new replacing part or book an appointment with the company's support. In this way, and thanks to the modular design, the useful life of the product can be significantly extended.

8.5. Prototyping

The author had access to the 3D printer model F370 from Stratasys. The final model of the prototype was highly dependent on printer's capability of maximum build size which is 350 (width) $\times$ 350 (height) $\times$ 250 (depth) millimetres. The Solidworks model parts of the planter were converted into an STL file which is a standard file format for almost all 3D printers. The pots and lids were printed with Stratasys Objet30 Prime. Both printers use standard ABS material for production. Nonetheless, the nozzle of the F370 prints thicker layers with microscopic holes or cavities in the surface. This makes the final product porous, which is an issue with particularly the bottom part of the planter, where a tank is meant to hold the nutritional liquid mix. Thus, the main parts were sanded, and subsequently, a water-resistant coating was applied. The Objet30 can only small print small parts, however, those are already waterproof as the printing technique is very precise and leaves no porous surfaces. The top upper part (i.e. lid) was produced by an external company as it was necessary to make the lid partly translucent with a white colouring for the light to diffuse through. This is due to the fact that the LED lights are positioned just right under the lid. The appearance is similar to frosted glass or a diffusing plastic material used for modern LED chandeliers or lamps (Figure 23).



Figure 23 – Light dispersion effect

When all the parts were printed, and electronics collected, the prototype was assembled (Appendix 21). Unfortunately, there was not enough time to fully wire it and carry out further tests of its functionality and establish efficiency metrics.

9. Nutrient management

The success or failure of the system primarily depends on the strict nutrient management. It is important to supply accurate essential nutrient on time with required concentration. Therefore, it is important to manipulate and adjust the concentration and pH level of the water nutrient solution. Monitoring the plant's nutritional intake is vital; if the nutrient levels of the solution are too high the plant could fail due to the root burn. Also, its chances of survival are highly diminished if it does not receive enough nutrients. "Low EC results in slow growing plants with disorders such as leaf yellowing, whereas excessively high EC can lead to toxicity and poor plant growth" (AEssense, 2016). The main necessary nutrient components of any fertilizer are Nitrogen (N), Phosphorous (P) and Potassium (K) also known as the NPK ratio. The NPK measurement was invented as a way to measure the quality and concentration of the three most vital nutrients within a particular solid or liquid fertilizer (Science in Hydroponics, 2010). These three numbers represent the percentage composition by weight of any given fertilizer, telling us its percentage composition of N as nitrogen, K as K_2O and P as P_2O_5 . For example, NPK ratio 1-0.76-2 of a fertilizer mix consists of 1% nitrogen, 0.76% phosphorus and 2% potassium. In addition, most plants require small concentrations of: Boron (B), Calcium (Ca), Chlorine (Cl), Magnesium (Mg), Iron (Fe), Sulphur (S), Copper (Cu), Manganese (Mn), Molybdenum (Mo) and Zinc (Zn).

One way to measure the concentrations of salts in a liquid mix is to use electric conductivity (EC) or total dissolved salts (TDS) meter. There is also another way to measure the salt concentrations, and that is to conduct a gravimetric analysis (Environmental Express, 2013). As the liquid is vaporised, the remaining residue is dried and weighed. However, this

method is time-consuming and cannot be utilized with immediate feedback as in comparison with EC/TDS sensor measuring technique. So, an experiment was carried out in a chemical lab at Brunel University under experienced chemistry supervision. The aim was to see the effect of different chemical components on the EC/TDS readings. The key hypothesis of this test was that a correlation exists between the concentration of dissolved solid particles and EC/TDS readings of the meter/sensor. In theory, the more salts are dissolved in a solution, the higher is the value of electrical conductivity. All meters and sensors were calibrated prior to the start of the test. EC/TDS meters were calibrated against 1413 uS/cm calibration solution from Hanna Instruments and pH meters were calibrated against standard pH4 and pH7 solutions available in the lab. Three different hand-held EC/TDS meters (Figure 24) and an Arduino-connected DFRobot TDS meter (Figure 25) were used for the tests. pH measurements were also recorded with the tester from Hanna Instruments and BNC Probe connected to an Arduino-compatible signal transmitter board. Additionally, a bit older but well-functioning general-purpose pH meter Mettler DELTA 350 was accessible in the laboratory.



Figure 24 – Hanna HI-98129, Xiaomi Mi TDS meter and Xiaomi Mi Flora

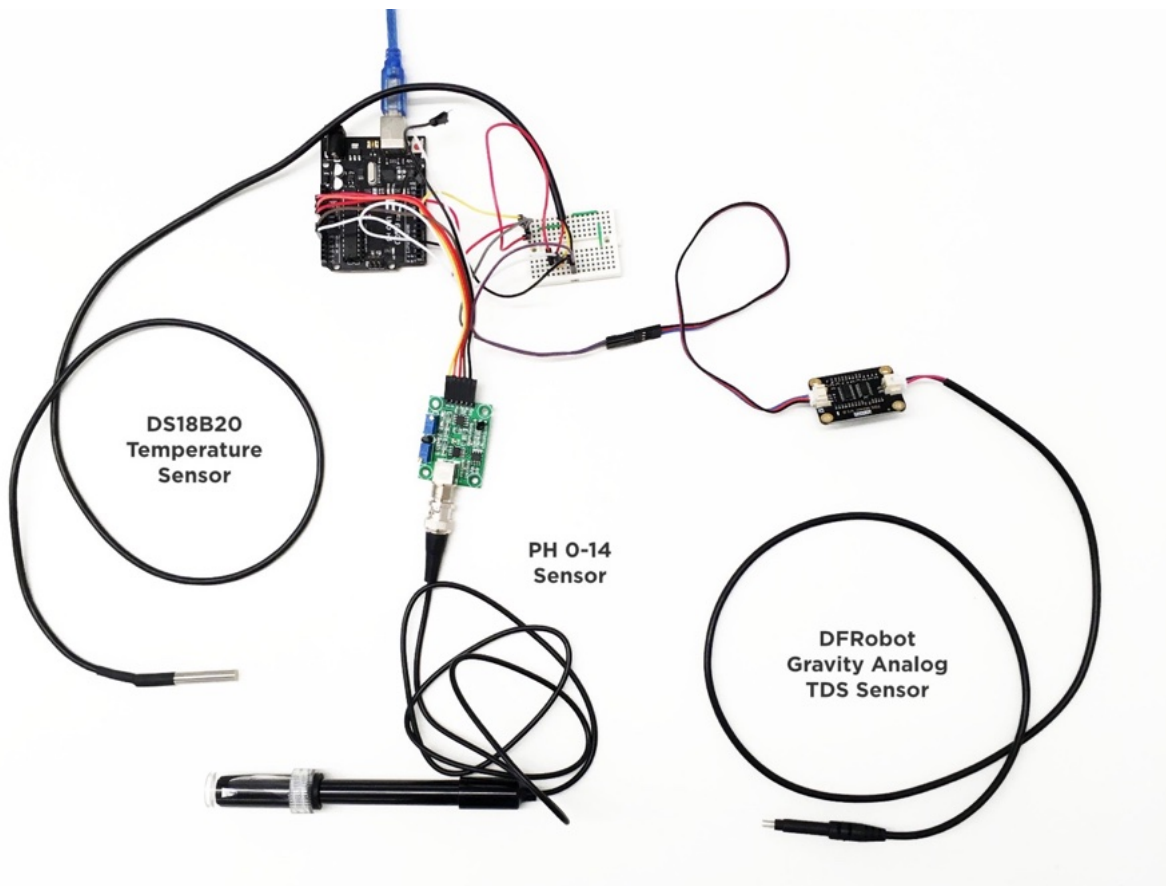


Figure 25 – Arduino-connected sensor modules used for the data analysis

Electronic meters of *total dissolved solids* (TDS) were utilized to measure the estimated concentration of ionic solids (i.e. salts) in the solution. TDS is measured in *parts per million* (ppm) where 1 ppm equals to 1 mg of the mineral compound in one litre of the solution (i.e. 1 mg in 1,000,000 ml). It is the total weight of all solids in a unit amount of solution (Covey, 2012). Pure water is meant to have conductivity measurement of zero due to the absence of salts. It is important to note that ppm readings do not always accurately represent the actual weight of the ionic solids dissolved in the water solution as TDS is simply a conversion of the *electric conductivity* (EC) measured in milliSiemens (mS) or microSiemens (μ S). Whereas the “true” value of the TDS include ionic (i.e. anions and cations) as well as non-ionic solids which do not contribute to the conductivity. Moreover, different ions have different electric conductivity when dissolved in water, ergo the contributions to the resulting conductivity are not same for all (AEssense, 2016). Previous research cautions that the correlation between TDS

and EC is not usually linear, even for simple salts (Thomas, 1986). The so-called conductive mobility of ionic species (i.e. Na^+ , K^+ , Mg^{2+} , Ca^{2+} , Cl^- , HCO_3^- , NO_3^- , SO_4^{2-} and CO_3^{2-}) is variable (Corwin and Yemoto, 2017). For instance, univalent cations such as Na^+ are more mobile than multivalent ions such as Ca^{2+} and Al^{3+} “It depends on the activity of specific dissolved ions, average activity of all ions in the liquid, and ionic strength” (Rusydi, 2018). There are some solids or salts that do not yield ions to change the conductivity of the solution for example (Covey, 2012). Therefore, the EC cannot always be used as a final mean for determining the overall value of the solids in the solution (i.e. TDS). Most TDS meters simply multiply the EC version with a pre-determined conversion factor k . Therefore, the challenge lays in the selection of the correct value for the conversion factor. There is an ongoing research into what is the optimal number (Corwin and Yemoto, 2017, Rusydi, 2018). In theory, every ion has its own specific conductivity coefficient (Lenntech, 2018), and as mentioned before, the ions interact with each other, hence changing the overall conductivity of the water quite unpredictably. This is due to the complex relations between solute and solvent. Going more in depth with detailed calculations of ion-specific conductivity and appropriate EC/TDS conversion factor would be beneficial but not crucial for the aims of this paper. Nevertheless, research suggests that the approximate value of EC/TDS conversion factor should be around 0.5 to 0.75 (Corwin and Yemoto, 2017, Rusydi, 2018). Most commercially available EC/TDS meters for consumers have a pre-set factor of 0.5; which is an estimated relationship of EC/TDS for sodium chloride (Aqion, 2018; Costa, 2018).

$$TDS = k * EC = 0.5 * 100 (\mu\text{S}/\text{cm}) = 50 (\text{mg}/\text{l})$$

$$2 \mu\text{S}/\text{cm} \approx 1 \text{ ppm } (\text{mg}/\text{L})$$

Firstly, 'ultrapure' or Milli-Q water available in the lab was tested. As expected it had zero conductivity as this type of water is free of any salts. It was used as the main solvent for

all the mixtures, so the resulting conductivity values would comprise of only the dissolved minerals. Next, regular tap water showed EC/TDS values with fairly consistent values around 340ppm. There was already a difference between the values obtained from each meter. Specifically, Xiaomi Mi Flora meter displayed about 1/3 higher value compared to other meters. The rest of the tests comprised of solutions mixed with Milli-Q water, so any EC/TDS values would only account for the added mineral compounds.



Figure 26 – Tested nutrient compounds (Minerals Water, 2018)

Separate nutritional compounds and a pre-mixed fertilizer from Formulex (Figure 26) of different volumes were dissolved in the water to test the how it affects the readings of EC/TDS as well as pH (i.e. alkalinity or acidity of the solution). 0.1g (100mg) and 0.01g (10mg) of a mineral solute were added to 100ml of a solvent Mili-Q water. The full analysis is provided in the Appendix 22. The data obtained from HI-98129 correspond very closely with the ones from Xiaomi Mi TDS meter and data from Xiaomi Mi Flora EC meter correspond with the ones from DFRobot EC Sensor. HI-98129 is likely the most precise out of all used meters since it is a professional lab tool. The calibration of this meter required a specific calibration solution from Hannah Instruments. The advantage of the EC/TDS from DFRobot lies in its configurability as the code can be modified and the probe calibrated. Moreover, Mi Flora is a Bluetooth-connected sensor for soil rather than liquid solution analysis. Due to this

fact it may have different internal algorithm, calculations and/or technology and so the resulting EC values too. A solution with sodium chloride—a regular table salt—confirmed the assumption about 0.5 coefficient between EC and TDS. The readings also confirm that 1mg of NaCl equals 1ppm in one litre of water. This notion does not, however, apply to other dissolved mineral compounds as previously mentioned. For instance, the last measured nutrition mix consisted of 0.349g of $\text{CH}_4\text{N}_2\text{O}$, 0.328g of $\text{NH}_4\text{H}_2\text{PO}_4$ and 0.192g of KSO_4 in order to achieve consistency of 200ppm of N, 100ppm of P and 200ppm of K in 1 litre of water. The compounds were prepared with precise weights, yet the sensor readings did not show 300ppm (100ppm P + 200ppm K; N is non-conductive) as would be expected. The results were close but not as consistent between the meters.

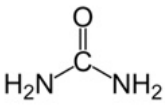
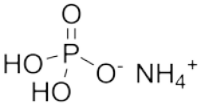
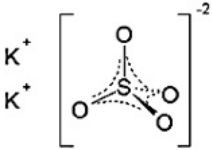
Name	Urea	Monoammonium Phosphate	Potassium Sulphate	Formulex (universal profile)
Chemical formula	$\text{CH}_4\text{N}_2\text{O}$ 	$\text{NH}_4\text{H}_2\text{PO}_4$ 	K_2SO_4 	2.3 : 0.9 : 3.4 1 litre 1.096 kg Analysis % w/v Nitrogen (N) 2.29 Nitrate nitrogen 2.14 Ammoniacal nitrogen 0.15 Phosphorus pentoxide (P_2O_5) water soluble (P) 0.87 (0.38) Potassium oxide (K_2O) (K) 3.36 (2.79) Calcium oxide (CaO) (Ca) 1.85 (1.32) Copper (Cu) chelated by EDTA 0.002 Iron (Fe) chelated by EDTA 0.040 Manganese (Mn) chelated by EDTA 0.010 Molybdenum (Mo) 0.001 Zinc (Zn) chelated by EDTA 0.0025
CAS number	57-13-16	7722-76-1	7778-80-5	
NPK	46-0-0	12-61-0	0-0-52	

Table 1 – Specifications of tested fertilizer compounds

It is enough to know that mix of individual chemical compounds of the fertilizer provides a certain resulting EC/TDS/pH value. In this way, the correct ratio of the fertilizer compounds can be adjusted to reach the desired level. The DFRobot sensor can be programmed to estimate the salt levels more precisely if the solution type is known. Also, the composition of the water is unknown and differs in each location. Certain assumptions like this one can be modified in the code thereafter too. However, the TDS does not tell the composition of the mix, it is only a mere estimate of the dissolved solids. Currently, there are no meters that would

measure the concentration of individual ions reliably. 2% per degree Celsius is an approximate correction value for temperature as ion mobility and so conductivity increases with the temperature (Mosaic Industries, 2016). The EC is likely to substantially increase in higher solution concentrations, that is another notion should be considered in the Aéropot sensor measurements and optimization of the nutrient mix (Table 2). Nevertheless, it is upon the user to make sure the ratios of the individual chemical compounds, namely NPK, are mixed and sustained in the correct ratio, or at least in a given range.

Concentration (%)	0.1	0.2	0.3	1.0	5.0
EC (mS/cm)	0.86	1.66	2.5	7.4	27.8

Table 2 – Changing conductivity of Mono-Ammonium Phosphate in different concentrations (Haifa Chemicals, 2018)

The author learnt how to carry out the calculations in the process of preparing own nutrient mix. The author also gathered an array of resources, from contacts to fertilizer suppliers to practical information about optimal mineral concentrations for a different type of plants (Figure 27). To conclude, the obtained data can support the understanding of the readings for the further testing and correct functioning of the Aéropot prototype.

	Steiner (1984)	Cooper (1988)	Hewitt (1966)	Hoagland and Arnon (1938)
Nutrient	mg L ⁻¹			
N	168	200–236	168	210
P	31	60	41	31
K	273	300	156	234
Ca	180	170–185	160	160
Mg	48	50	36	34
S	336	68	48	64
Fe	2–4	12	2.8	2.5
Cu	0.02	0.1	0.064	0.02
Zn	0.11	0.1	0.065	0.05
Mn	0.62	2.0	0.54	0.5
B	0.44	0.3	0.54	0.5
Mo	Not considered	0.2	0.04	0.01

Figure 27 – Optimal mineral elements concentration for hydroponic systems by different authors (Lakhiar et al., 2018)

10. Business plan

10.1. Mission

Aéropot is the world's first smart and functional ultrasonic system for effective plant cultivation in a soilless environment. It is a full system, including a physical planter, that maintains an optimum growing environment with a press of a button. It integrates seamlessly with a web and smartphone app. Aéropot's mission is to create a product that brings a feeling of joy through the cultivation of plants, encourages learning, connection and appreciation for nature and increases production of clean food in urban settings. Combination of highly-developed functionality and clean natural forms induces a creation of a durable bond between the user and product. It engages people's curiosity and interest in cultivation of plants. The mission statement represents the company's efforts in addressing business opportunities in the soilless cultivation agriculture and smart technology and contribute towards diminishing the negative effects of environmental and societal issues in the 21st century. For instance, addressing the issue of scarcity of water and land that is required to perform agriculture activities may alleviate associated social problems inflicted by such material insecurities. Furthermore, Aéropot's distinctive product philosophy is fostered by three fundamental pillars: useful function, clean design language, and intuitive control and monitoring. The product is useful through effectively growing plants in a controlled environment, achieving higher yields and shorter growing cycle. The visually pleasant and simple outlook is realised through careful selection of suitable design elements and principles with attribution of precise mathematical definition. Thanks to this consideration, the product fits into any modern household or indoor environment. The top priority is placed on user experience, any interaction with the product—including planter and an app—should feel as natural and easily understandable as possible. Nowadays, consumers expect real-time feedback from electronic devices and products that are ready to be used right out of the box. There is a shift towards automation and the product is

expected to perform otherwise repetitive tasks and automatically act upon sudden changes in the system.

“The aim of product design is to make the technology so simple and easy to use that anyone—even a child—can use it” (Dholakia, 2012).

10.2. Competitor analysis

There is no official analysis or report of the -ponic market to this date. Therefore, all known companies, that commercially offer a product in this market, were investigated. A detailed competitor analysis of hydroponic and aeroponic device market is represented in the Appendix 23. In the paragraphs below are concluding remarks that stem out of the analysis:

- The market is fairly concentrated in terms of hydroponic/aeroponic device solutions.
- Not many ultraponic devices have penetrated the market yet.
- Most of them are hydroponic-based systems (pump or passive).
- The market segment (-ponic home solutions) can be divided into two in terms of purpose and appearance: aesthetically pleasing ‘gadgets’ for the cultivation of plants on a small scale and more ‘bulky’ systems for industrial-scale growing and experimentation.
- Prices vary from £50.00 to £200.00.
- Only two out of thirteen (15%) analyzed systems use a smartphone app.
- The common message in the value proposition is the convenience and automation of growing plants at home without much effort and technical understanding.
- The first half of the companies built up the brand and marketing strategy. Communication through social media channels and blog articles has been crucial for them.
- The second half (mostly manufacturers themselves) do not use social media, simply selling their products through a network of retailers.

- There are many ‘nobrand’ products from Chinese manufacturers that are offered on marketplaces such as Amazon or eBay.
- On average, the devices can grow around 8 plants. This varies according to the size and purpose of the device.
- Supplemental lighting must be added to the planter in most cases.
- Almost none of them provides any sort of educational materials to teach the user about the underlying processes.
- There is no comparison to other competitors in terms of performance.
- In general, companies do not provide any technical data or results of tests on the effectiveness of the device.
- No smart/truly automated product on the market

To be absolutely transparent and true with the analysis, there is one more student concept ‘Pod fogponic garden system’ by Greenfingers (Figure 26, left). Another fogponics device known as ‘FogBox’ is a desktop aeroponics system released exclusively through a crowdfunding campaign (middle). Those are purely aeroponic devices, sometimes also called ‘fogponics’, which utilize the ultrasonic misting device. Lastly, a French company called GUS claims to be offering the “world’s first ultraponic growing tray”. Although their product seems to function well, they were not able to raise enough funds through Kickstarter campaign. The price of one unit was priced around €549.00 which was fairly high in comparison to other -ponic products on the market. None of the aforementioned products are commercially available [anymore].



Figure 1 – Fully developed ultrasonic growing planters (Designboom, 2012; GroBox, 2017; GUS, 2018)

An American company SproutsIO is possibly the closest competitor (Figure 27). It is a holistic automated indoor growing system that is, however, still in development. “It includes everything you need to grow with no hidden add-ons all in one thoughtfully designed experience. The device is equipped with WiFi connectivity, environmental sensors for monitoring both plant and ambient conditions, camera, high efficiency LED lighting, and an electronic mister” (SproutsIO, 2018).



Figure 2 – SproutsIO (2018)

10.3. Business model canvas

A full table-type business model canvas was developed, containing all the necessary details in each section regarding firm's/product's value proposition, infrastructure, customers, finances, etc. (Appendix 24).

<i>Strengths</i>	<i>Weaknesses</i>
- Result efficiency - Innovative design (hardware/software) - Intuitive control - Variety of uses	- Small-scale - Low volume
<i>Opportunities</i>	<i>Threats</i>
- Market gap (quality/value for the price) - Ultrasonic misting ('ultraponics') - Attractive modern appearance (+ light effect) - Smart/IoT product	- Development (time and resources) - Reliability (requires testing) - Intellectual property claims

Table 3 – SWOT analysis

The benefits of soilless plant cultivation technique -ponic techniques were already described in sufficient depth. Coming back to the competitor analysis, there is certainly a gap in the market, even though it is a fairly concentrated market. Moreover, a quick SWOT analysis was conducted (Table 3) to identify the combination of factors and other advantages of the product that can create a strong value proposition for Aérospot product and its business model:

- Superior convenience, energy and resource saving properties
- Multiple functions and ways to use
- Effective use of space (i.e. fit more plants, a smart positioning of electronics)
- High-tech, sleek design
- Customizable/modular design (e.g. upper/bottom parts can be changed/combined)
- Unobtrusive design (i.e. LED implemented in the case, magnetic opening, no cables, etc.)
- High-quality materials and build structure

- Intuitive monitoring and control
- The retail price can be less or equivalent to the other products available on the market, around £80 to £100
- Offer different versions (e.g. minimal, essential, smart)
- Strong brand identity (i.e. design language, mission, brand values, etc.)

Cost analysis of the product was also carried out (Appendix 25). The overall cost of the electronics for building the prototype came about £50. The electronic components were bought as a single product, so the price would substantially drop if it was bought in a bulk. If the product was manufactured in a serial production, the cost would also go down too. The price analysis excludes the making of the chassis as it was 3D printed, which is a technique that is normally very expensive. Eventually, the casing, individual pots and other plastic parts should be produced utilizing an injection moulding technique. The parts could be made out of 100% recycled BPA-free polypropylene (PP) and/or polyethylene (PE) plastic. PP and PE were chosen for their stiffness and outlook properties. The fertilizers are not a considerable cost for the customer (Appendix 26).

Just to expand a bit on the revenue streams, the Aéropot planter would be sold as a whole primarily through the web/ecommerce shop which has the benefit of full control and no fees. To reach more customers, another selling channel through online marketplace platforms such as eBay or Amazon would be established. Additionally, the user would be able to order new parts through the app in a case of malfunction. The app would guide the user step-by-step through the process of repair. Similarly, there would be a variety of subscription plans that would deliver a ready-to-use package of all necessary ingredients to the user. For instance, a set of seeds with specific fertilizer would be delivered and simultaneously the timings and light intensities of the planter would adjust according to their requirements. A user would be enabled to opt-in for this subscription-type mode to receive support and feedback at every step, which

would be suitable for beginners. It would require minimal expertise and effort, thus that would attract a wider consumer base. Nevertheless, at the same time, the system would allow for experimentation mode where a more experienced user could adjust any parameter.

10.4. Target market

“Urban gardening: consumers are trying to recreate a gardening experience inside their homes, boosting sales of indoor plants. A steady focus on health and wellness bolsters demand for grow-your-own vegetables and fruit. Simultaneously, consumers are increasingly viewing gardens as outdoor entertainment areas and are looking to transform balconies into green living rooms.”

There is a large community of hydroponic enthusiast all over the world, specifically in the western hemisphere. Besides this particular cohort, not many people really know what hydroponics nor aeroponics stand for. The competitor analysis revealed that those companies fail to reach a higher number of consumer audience. One hypothesis may have to do with the fact that consumers are not being educated about the idea and benefits of the soilless plant cultivation. However, this is likely to change as the technology is still developing. So, it is only a matter of time before the interest and engagement grow among the general public, health community and specifically urban agriculture investors. The target groups stem from the literature review and general observations of the market (Table 4). The product is interesting for office spaces as a decoration with minimal maintenance and aesthetically pleasing design. It is an educational tool for children that introduces them to responsible and environmentally friendly plant cultivation. It also fosters the learning and understanding of underlying processes in nature and novel developments in the field of smart technology. When the Aérospot is

released on the market, it would be the first European smart ultrasonic system available for wider consumer base, rather than industrial market only.

<i>Millennials</i>	<i>Baby boomers</i>	<i>Offices</i>	<i>Children</i>
- 37% of Millennials are growing herbs indoors - Want to express themselves in unique ways - Appreciate outdoor, natural spaces and plants for their contributions to mental and physical health - Want organic/green solutions to soil health and fertility - Appreciate a less-is-more aesthetic	- Gardening remains an activity for older generations - Looking for convenience and ease of use	- Aesthetically pleasing ‘artisan’ natural object - Improve working space conditions - Employees prefer offices with a reasonable number of plants	- Reconnecting with nature in 21st century - Educational purposes - Children connected with nature are more likely to be interested in their environment and will strive to enjoy it and preserve it

Table 4 – Market segmentation (RSPB, 2013; Petersen, E. 2016; Gelsthorpe, 2017; Euromonitor International, 2018a)

10.5. Future plans

Firstly, the author will be pursuing professional development of the MSc course (i.e. the ‘Prototype Development Pathway’) at Brunel University, where most of the product development and experiments took place (Appendix 27), to further develop the business plan and marketable version of the product. The next steps involve: fully finishing the prototype, so it can be utilized for presentation purposes; establishing connectivity channels between the device modules and cloud-based services; development of the alpha version the web-based app (eventually compatible with iOS and Android systems); followed by electronic tests for waterproof and connectivity efficiency as well as tests of plant growth and nutrient absorption efficiency (Figure 28).

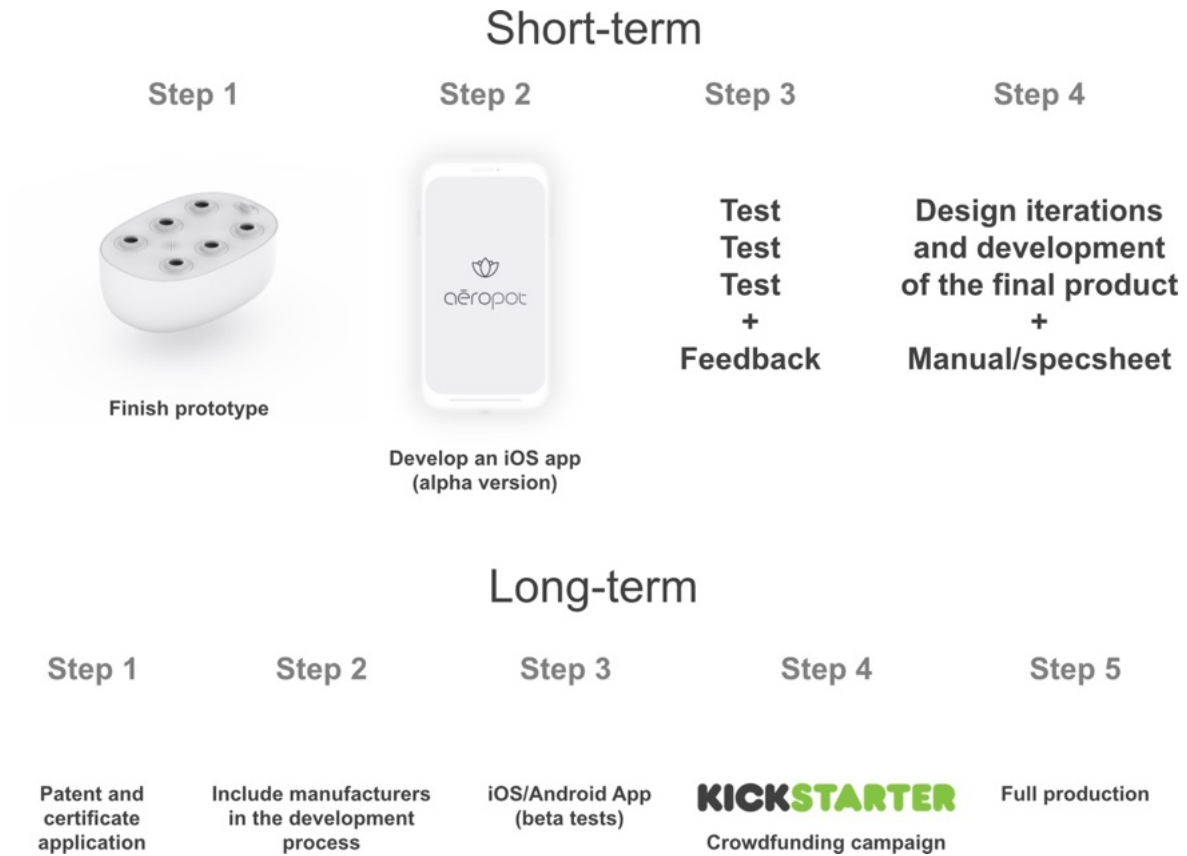


Figure 3 – Future plan with defined objectives

Later, the prototype requires testing in terms of nutrient absorption and growth efficiency as well as electronic waterproof and connectivity properties. A showcase session should be carried out where the participants are free to use and play with the product in return for constructive feedback. It would be beneficial to construct a survey or a semi-structured interview to gain a full understanding of relationship and patterns in terms of gardening habits, with consideration of different age cohorts. It would be ideal to have a marketable version of the product in the end with all design and engineering iterations incorporated and ready for manufacturing. Connections with suppliers and manufacturers will be established at the same time as selling channels (i.e. ecommerce shop, retailers and business partners).

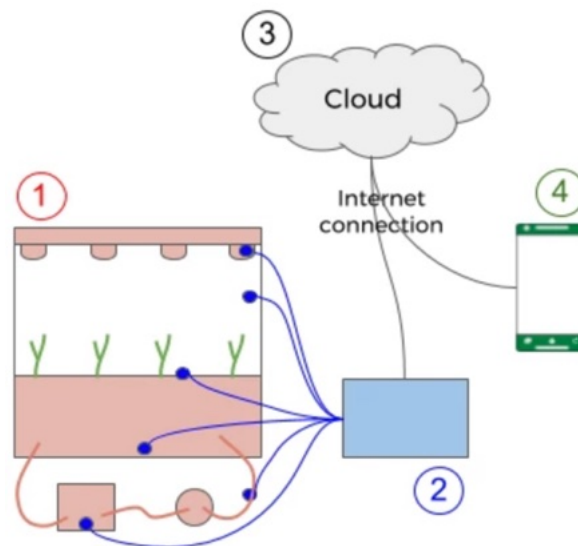


Figure 4 – Simplified diagram representation of the smart-ponic system (Growstack, 2018)

The final system consists of the planter where electronics, sensors and actuators are connected to a microcontroller that communicates through a cloud service with the smartphone app (Figure 29). In the future, the quantitative results from the sensor tests should be further compared to the chemical reaction equations; to see if the measured values correspond with the calculations to some degree. Future research and more literature review should be done in the area of salts and associated ions. Specifically, what is their influence on the conductivity and how to calculate it. This would help to include calculations into an algorithm in the TDS meter that would measure the levels more precisely if the information about the weight added salts was considered. Practical tests should be carried out regarding the optimal droplet sizes and flow rates for different type of plants. There is still not a conclusive evidence for this particular subject (Hikosaka, 2016, Lakhia et al., 2018). Different frequency settings of the ultrasonic mister should be tested to see the effect on the generated droplet size. Combination of ultrasonic foggers along with the low-pressure aeroponic design in the same system should be tested too. This is still very much a grey area, where most of the information comes from hobbyists posting their achievements online, rather than concrete and reliable research with valid conclusions. The end goal is to find a combination of variables and conditions that create an optimal

environment for the plant to thrive in and reach optimum levels of appearance, yield and flavour.

One disadvantage of the current design is the fact that the ultrasonic atomizer has a height of around 2.5cm, so any nutrient liquid mix under this level cannot be vaporized. At the same time, the dissolved salts are likely to settle at the bottom of the planter and the mister cannot atomize anything that not in contact with the piezoceramic disk. This is one of the major design concerns that needs to improve in the future versions of the planter. Another challenge is pathogen control (i.e. growth of bacteria, fungi and/or algae). Prolonged use test of the planter will show if this is a serious issue or not. Some strategies to deal with pathogens may include: keeping optimum temperature, regular maintenance, anti-bacterial substrate, and UV light inside the canopy. Lastly, a detailed inspirational moodboard for the control app was built (Appendix 28); it consists of collection of app screen examples, conceptual UIs, graphs, and others. It will be subsequently utilized as base for the development of Aérospot app's UI/UX. The Aérospot's final user interface will be constructed as combination between specific requirements of the Aérospot system and best-case scenarios, structures and features stemming from the moodboard analysis.

11. Discussion

This paper maps the research and development of a smart ultrasonic system for indoor plant cultivation named Aérospot. This is a multidisciplinary project spanning across a variety of fields such as entrepreneurship, design, engineering, information technology and environmental sciences, with emphasis on practical outcomes. Most of the set objectives of this study were achieved. The paper explored the theoretical background, market trends and soilless plant cultivation technology. Besides a business plan of the company, as simple branding strategy/visual guideline with a few elements such as logo, colour pallet and company fonts were developed. There have been multiple practical outcomes such as the 3D print of the

planter, a collection of all electronic components, wiring schematic, business plan, competitor analysis, roadmap, conditional logic (i.e. protocols), lab data tests and presentation. The best practices of -ponic systems were thoroughly studied from the hands-on experiences of hobbyists and academic theoretical knowledge and experiments. The project accommodates the method of trial and error, where the mechanisms and technology are tested to see what works and what does not. In order for the project to be as close to reality as possible, the planter was designed according to the limitations of the currently available technology. The availability of the electronic components on the market was the most crucial factor that influenced the final design. The prototype was construed such a way to achieve the highest possible degree of effectivity. Many variables had to be considered for each electronic component such as positioning of the components, a shape of the planter and other more specific considerations. For example, in the process of choosing the correct light variables such as: colour, light intensity, size, shape, waterproof properties, etc. had to be considered. The analysis of the market for technology and available components showed what is actually possible. So instead of proposing a design concept without consideration of the current reality, all the suggested design and engineering features have their functionality proven. The final deliverable is a tested and functioning prototype with incorporated electronic components, desired design features and customizable control preferences. The final product is echoing trends in consumer electronics for more refined and intuitive products, answering the urgent calls for creating sustainable food sources and enabling people to independently grow healthy vegetables, fruits and other plants. The author gained experience with Arduino wiring and coding, even though he had a prior knowledge in of electrical engineering and coding languages such as C#. So, there was a certain learning curve. This also applies to conductivity related theories and calculations necessary for understanding the influence of fertilizer compounds. Both knowledge areas were necessary for the final construction of the planter,

correct internal wiring and algorithm function. The Aérospot project was submitted to the Hardware Acceleration Programme at Central Research Laboratory in Hayes and a Santander sponsored, ‘Why Walk when you could Run?’ competition and in both cases successfully chosen as a finalist for the business pitch (Appendix 29). The project proposal will be submitted to more competitions, accelerators and/or incubators in order to receive valuable feedback (to fine-tune the business plan and product) and potentially gain some investment for further development.

12. Conclusion

This report firstly examined the environmental theories, gardening market and alternative indoor systems to cultivate plants. Multiple different methods of analysis and frameworks were utilized to evaluate the desirability, feasibility and viability of the product while following the generic model of the triple bottom line (i.e. economic, environmental and social dimensions of development and results). A multifunctional aeroponic device solution to effectively and holistically tackle the 21st-century challenges from sustainability, design and business perspective was proposed. Aérospot is an intuitive and affordable tool for ecological education that is empowering individuals to get true data and understanding about their gardening efforts. It can cultivate plants with a small amount of water and without soil, which has the potential to address the issues of land and water scarcity and contribute to the efforts of ending world hunger and poverty. This paper laid a solid ground for the realization of the marketable product. The gathered knowledge here can be referred back to during the process of forthcoming development.

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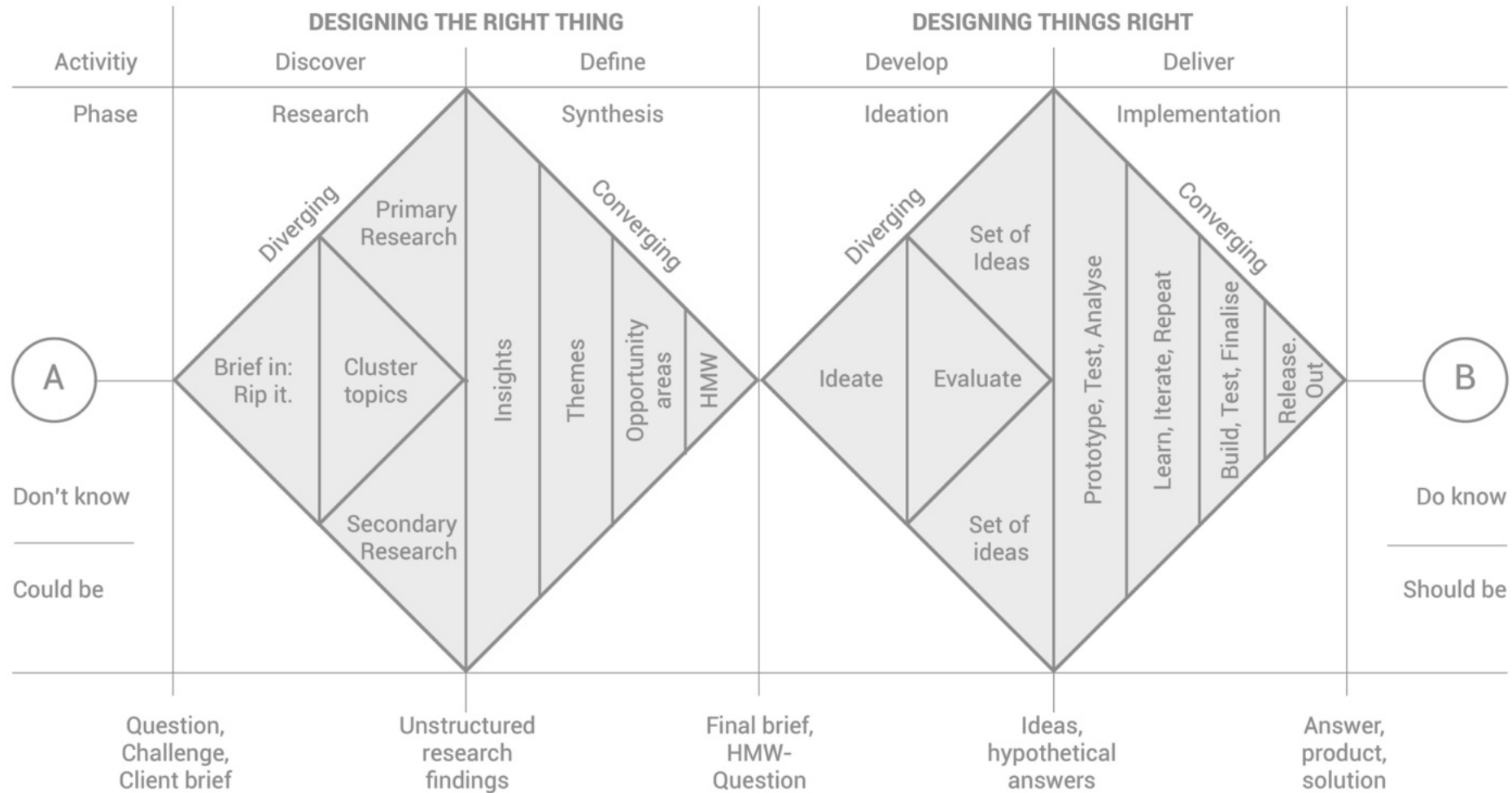
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Appendices



Appendix 1 – Double Diamond design process (Nessler, 2016)

<i>Benefit</i>	<i>Description</i>	<i>Examples</i>
Psychological well-being	Positive effect on mental processes	Increased self-esteem
		Improved mood
		Reduced anger/frustration
		Psychological well-being
		Reduced anxiety
		Improved behaviour
Cognitive	Positive effect on cognitive ability or function	Attentional restoration
		Reduced mental fatigue
		Improved academic performance
		Education/learning opportunities
		Improved ability to perform tasks
		Improved cognitive function in children
		Improved productivity
Physiological	Positive effect on physical function and/or physical health	Stress reduction
		Reduced blood pressure
		Reduced cortisol levels
		Reduced headaches
		Reduced mortality rates from circulatory disease
		Faster healing
		Addiction recovery
		Perceived health/well-being

		Reduced cardiovascular, respiratory disease and long-term illness
		Reduced occurrence of illness
Social	Positive social effect at an individual, community or national scale	Facilitated social interaction
		Enables social empowerment
		Reduced crime rates
		Reduced violence
		Enables interracial interaction
		Social cohesion
		Social support
Spiritual	Positive effect on individual religious pursuits or spiritual well being	Increased inspiration
		Increased spiritual well-being
Tangible	Material goods that an individual can accrue for wealth or possession	Food supply
		Money

Appendix 2 – Typology of the benefits of interacting with nature (Keniger et al., 2013)

Parameters	Hydroponic system				Aeroponics
	Media soilless system		Nutrient solution system		
	Open	Closed	Open	Closed	
% Irrigation water saving	80	85	85	90	95
% Fertilizer saving	55	80	68	85	85
% Productivity increase	100	150	200	250	300
% Water productivity	1000	1600	2000	3500	8000

Appendix 3 – Quantitative differences between hydroponic and aeroponic systems (Al Shrouf, 2017)

Hyperconnectivity demands an omnichannel response

- Rising smartphone possession and internet access continue to boost demand for online sales in home improvement and gardening.
- Players are responding through augmented reality apps, which facilitate brand awareness.
- Retailers and manufacturers should be thinking in omnichannel, not multichannel terms.

Turn-key solutions to tempt Do-It-For-Me consumers

- Do-It-For-Me (DIFM) preferences are more common amongst millennials who lack basic DIY knowledge, however, DIFM behaviour is also growing amongst older baby boomers who face mobility issues.
- Turn-key solutions, including design consultations, free delivery or other similar services, attract consumers looking for value-adding interactions with home improvement and gardening players.

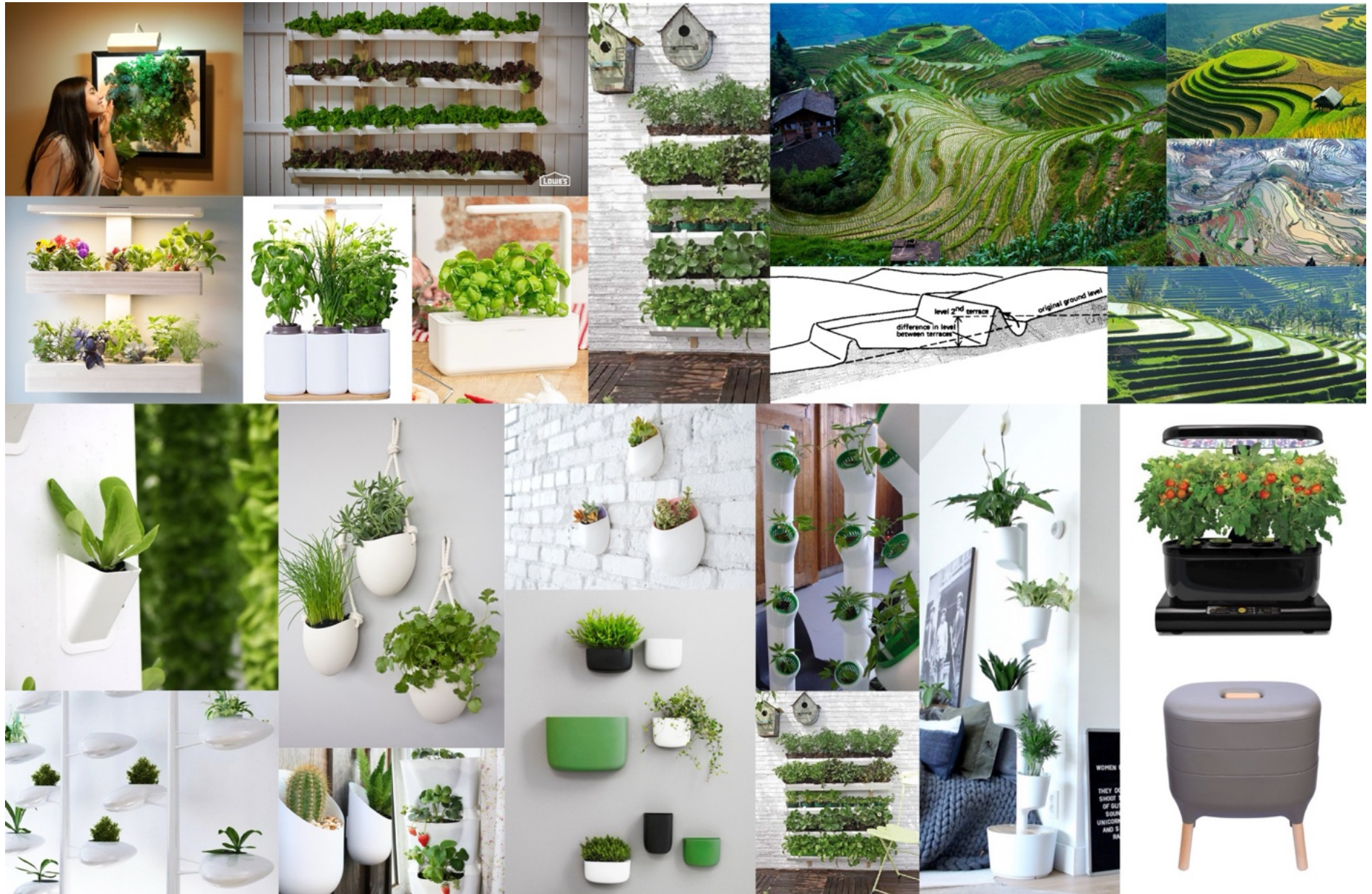
Urbanisation creates indoor gardening opportunities

- Urbanisation in both developed and developing markets results in more households living in apartments.
- Apartment living drives demand for urban gardening, through rooftop gardens and indoor plants.
- Smart gardening kits and hydroponic systems are gaining popularity in response to health and wellness trends.

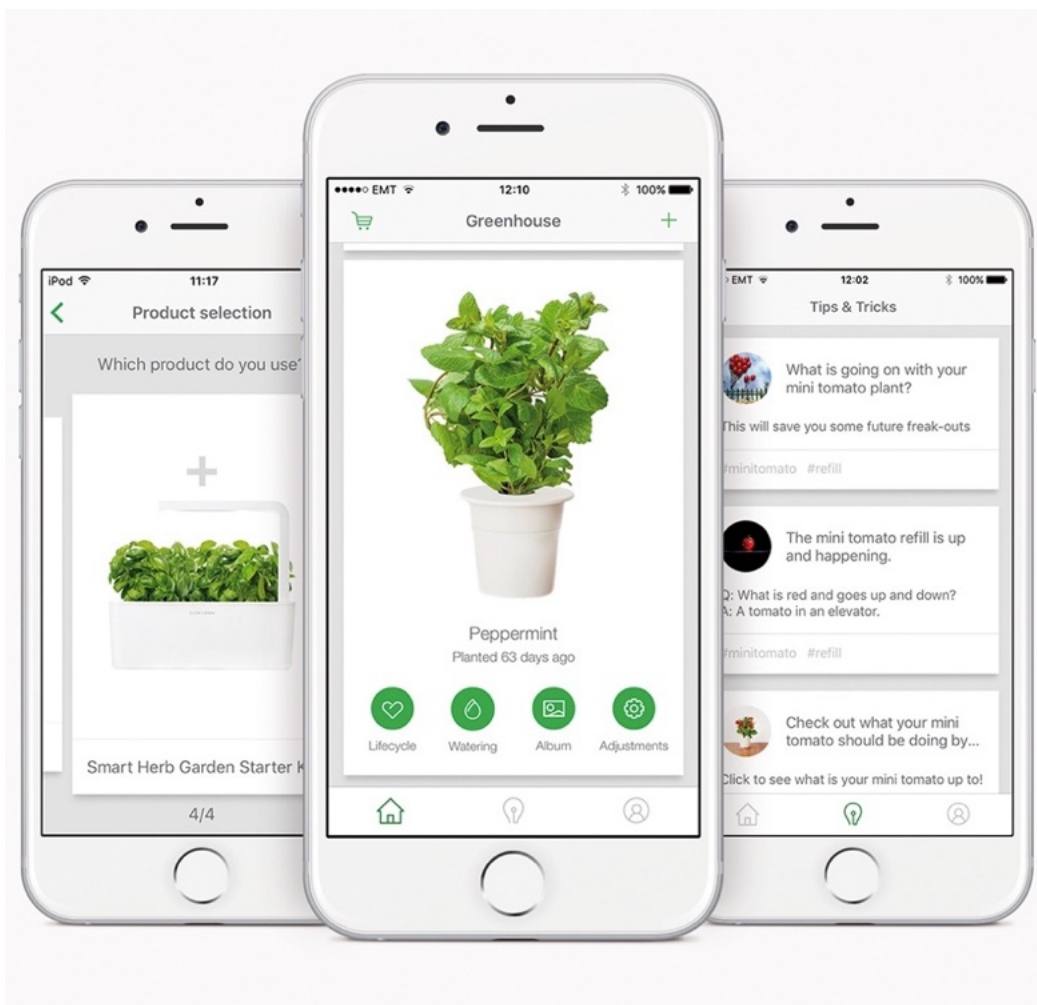
Smart home future is fast approaching

- The shift towards smart homes represents a major catalyst for change in the home improvement and gardening industry. Within the next five years smart home products are expected to become commonplace.
- Home improvement and gardening players that endorse smart home technology now will reap the rewards in the not-too-distant future.

Appendix 4 – Key trends in the home improvement and gardening industry (Euromonitor International, 2018b)



Appendix 6 – Moodboard



Appendix 7 – Click & Grow and AeroGarden smartphone apps

1 AÉROPOT

CONNECTING PEOPLE, NATURE AND TECHNOLOGY
AND MAKING GARDENING COOL AGAIN

Key Insights

POPULATION

67% of the world's population will live in cities by 2050

90% of time is spent indoors by most citizens in developed countries

63% of adults agree that gardening is a rewarding pastime

42% of British adults live in urban locations

15% of city dwellers have no outside space where they live

CONSUMER TRENDS

52% of all those who spent money on their homes in the last year said they did so to make it a more enjoyable place to live

78% of adults do not have any smart connected home devices in their homes

44% of those who do not already own smart devices show an interest in buying some

Germany and the United Kingdom are the top two countries in the world for smart home sales. In the Western Europe, smart home sales are accounting for a 30% value share

The UK has the most mature smart consumer spending is forecasted to grow by 28% and reach £7 billion by 2023

It is expected that companies will invest in products that allow users to spend time and money and that products that let them have a better experience of their homes will be the most successful. Smart home products such as IoT-enabled lighting, security systems, and smart appliances are expected to be the most successful.

Solutions

NATURE

Biofilia

"an innate and genetically determined affinity of human beings with the natural world"

As more people move from country areas to the city and as tend to build homes near the city centre becomes scarce, we're getting further and further away from nature. It turns out this isn't great for our health. The increase in prevalence of conditions such as obesity, attention disorders, and depression is partly due to a decrease in the degree we are exposed to nature. Even minor effects of adding plants can add up to a substantial decrease in the health burden on a global scale. Plants improve air quality and self-reported frequency, but indoor contact with them is also a very important factor.

Moreover, there is an ever-growing list of literature that suggests that nature elements must not be considered as an optional but an integral to satisfaction and well-being.

Looking out of a window into the street with a view of greenery will increase heart rate variability (HRV) and reduce blood pressure and stress levels.

Indoor plants

There are some very functional benefits that go far beyond the aesthetic and the positive psychological effect of being around plants.

- Increasing humidity
- Reducing carbon dioxide levels
- Reducing levels of pollutants (eg benzene and nitrogen dioxide)
- Reducing airborne dust levels
- Keeping air temperatures down

Plants that are good for indoor air quality: Spider Plant, Boston Fern, Snake Plant, Peace Lily, etc.

2 DESIGN

MINIMALISM

JAPANESE DESIGN

ESSENTIAL DESIGN

The concept of minimalism is to strip everything down to its essential quality and achieve simplicity. The idea is not completely without ornamentation, but that all parts, details, and colors are considered as reduced to a stage where no one can remove anything further to improve the design. The basic geometric forms, elements without decoration, simple materials and the repetitions of structures represent a sense of order and essential quality.

Brandenburg design is the forerunner of modernism. It is a design, producing products that are beautiful, accessible, and affordable and achieving design principles that use natural materials, clean lines, and straightforward construction.

Starting by Dieter Rams who laid down the rules for the modern minimalist in consumer appliances (eg. Braun products), with strong emphasis on user experience, and continuing up to Jonathan Van Noor who designs for Apple company.

Less, but better

Looking for "organic" forms that are inspired and possibly aligned with the natural systems. Organic minimalist style references the curves and shapes typically found in nature. It celebrates the harmonious balance between human habitation and the organic world.

TECHNOLOGY

Aeroponics

Aeroponics is the process of growing plants in an air or mist environment without the use of soil or an aggregate medium. When growing edible products (eg. pellets, mushrooms or cherry tomatoes), the effectiveness of aeroponic systems can be 70% higher in comparison to hydroponics (where plants are fully suspended in the liquid nutrient solution).

Advantages:

- Low risk of disease and pest infestation
- Plants grow up to 10x more quickly
- Low energy and water required
- Better exposure to oxygen
- Complete control
- It's the future

Smart Home

Smart homes serve users effectively through intercommunication between various computing devices embedded in everyday objects. In the ideal version of a smart future, all devices in smart homes communicate with one another seamlessly. Such automated buildings would have installed detection and control devices, such as air conditioning and heating, ventilation, lighting, hardware, security systems, etc.

Internet of Things (IoT): The world of things, with sensors and connected devices is entering every day.

MARKETING

Benchmarking

COMPETITOR ANALYSIS

COMPETITOR	PRICE	QUALITY	FEATURES	DESIGN	AVAILABILITY
ELITE & GROW	£100	★★★★	• Soil-based growing • "Smart" light • LED light • Water level sensor	★★★★	Available
ADRENALIN	£100	★★★★	• LED light • Water level sensor • "Smart" light • Water level sensor	★★★★	Available
AFRIKI	£80	★★★★	• Pump • Water level sensor • "Smart" light • Water level sensor	★★★★	Available
X-STREAM	£70	★★★★	• Pump • Water level sensor • "Smart" light • Water level sensor	★★★★	Available
ANDER	£130	★★★★	• Pump • Water level sensor • "Smart" light • Water level sensor	★★★★	Available

CONCLUSION: No comprehensive and unobtrusive smart aeroponic pot solution on the market at the moment.

3 The concept

AÉROPOT is a smart aeroponic device that fits into any environment thanks to its universal and elegant design. It aims to provide full, closely-monitored control over the plant's vitality. At the same time, it serves as a tool for ecological education. Combination of high functionality and clean natural form induces emotionally durable bond between the user and product. It encourages people to find appreciation and build respect for nature, technology and underlying scientific knowledge.

PRODUCT DESIGN

CONNECTING PEOPLE WITH NATURE

Classical aeroponic systems (see the left) are not made to be an object of aesthetically pleasing form, rather they are made for technical, higher-level purposes. However, the few aeroponic systems that are on the market are not very energy-efficient since they use water pumps, whereas AÉROPOT is using energy for only effective elements: growing medium, lighting, misting mechanism. "AÉRO" is connected to an internal logic board fitted with the app.

HOW IT WORKS:

- Step 1: The app is pre-installed on the smartphone. The right plants + nutrients + solution are added.
- Step 2: The app is filled with the plants + nutrients + solution.
- Step 3: The programme is started. The device automatically makes the route.

WAYS TO USE:

- Passive: Flower sprout, adding aesthetic value
- Productivity: Air purification, oxygen production
- Independence: Edible plants (vegetables, fruits, and herbs)
- Fun & Joy: "... simply helps to soothe mind and body after a long and stressful day"

USER INTERFACE

DESIGN FOR GARDEN INNOVATION 2018

3 Modules:

- 1. Monitor: Monitor plant health, water level, nutrient levels, etc.
- 2. Control: Control the system, adjust settings, etc.
- 3. Data: View historical data, trends, etc.

My AÉROPOT

Dashboard:

- Plant health: 95%
- Water level: 80%
- Nutrient levels: 90%
- Light intensity: 95%
- Temperature: 25°C
- Humidity: 60%

It is in your hands now...

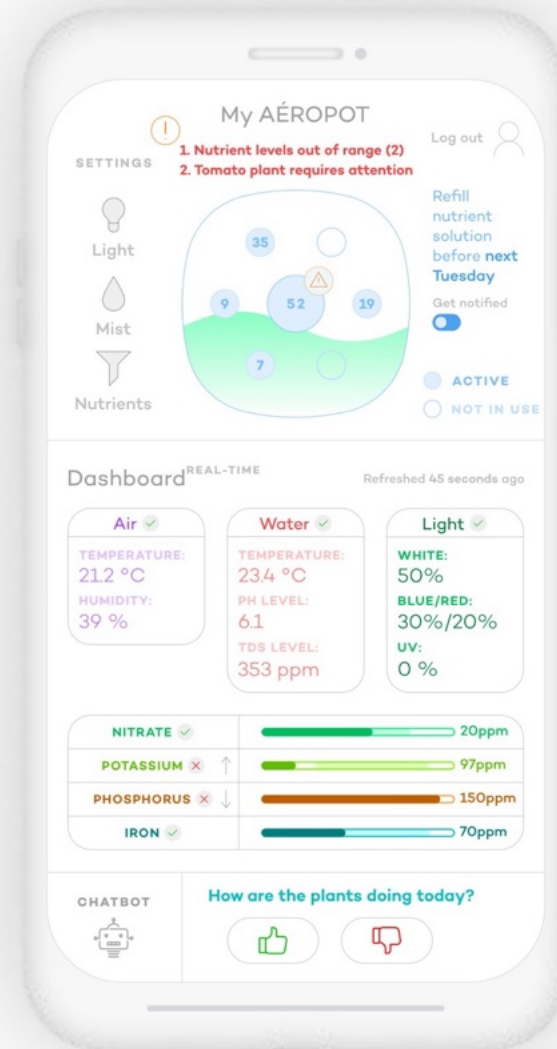
Aeroponic systems and growing plants in general require a lot of energy, time and resources. AÉROPOT allows for maintaining perfect levels of nurturing in the way which is the most critical criteria for the aeroponic system to be successful. Basic sensors are integrated into the chassis of the device, where optional modular sensors can be added anytime, which allows for detailed system monitoring and prevention.

AÉROPOT is an intuitive tool for ecological education that is empowering individuals to get true data and understanding of their gardening efforts.

Appendix 8 – Conceptual proposal of



Appendix 9 – 3D printed miniature of the Aéroport planter concept



Appendix 10 – The conceptual proposal of Aëropot's user app interface

<i>System</i>	- The materials and energy that is used for the production and manufacturing processes of the product. - On the other hand, the input energy is balanced out by the resulting output and specifically by increased efficiency of Aérospot device in comparison to conventional methods or other solutions on the market. For example, the amount of water normally necessary to grown certain number of plants in soil substantially decreases when grown with Aérospot. Moreover, the electronics incorporated in the device are much more energy efficient (i.e. ultrasonic misting instead of a pump or a microcontroller motherboard instead of a separate timer unit).
<i>Success</i>	- To sell a product that is pro-actively used by its users, adding value (i.e. knowledge, independence, function and/or pleasure) to their life and at the same time reducing the pressure onto the natural system.
<i>1.SP</i>	- The product is increasing concentrations of substances extracted from the Earth's crust since most of it is made out of synthetic plastics + electronics uses different metals and composites. - The nutrient liquid solution or fertilizer uses substances that could be made through energy-intensive processes. However, there are more environmentally friendly alternatives. - On the other hand, the liquid solution is almost always fully utilized/absorbed by the plants (difference from soil-based cultivation).
<i>2. SP</i>	- No toxins or other harmful substances released during the use of the product. - Some may be released during the production phase, but that is highly dependent on the manufacturer. - No waste, the product can be fully recycled by the end of its life. The chassis material can also be made out of only natural materials (e.g. bamboo). - In comparison to soil which has to be frequently fertilized and eventually must be discarded, the liquid nutrient solution completely absorbs into the plant.
<i>3. SP</i>	- In a "bigger Figure", the device is saving the use of natural resources (e.g. land or soil, water, etc.) and therefore avoiding degradation of the biosphere by physical means.
<i>4. SP</i>	- The device encourages healthy lifestyle by providing a way to grow organic plants for edibility, air purification, aesthetic or cultivation pleasure.
<i>5. SP</i>	- Aérospot is opening up possibilities of cultivating plants without a garden, connects and educates people about the underlying natural processes and gives its users access to real data about their gardening effects, therefore providing full control over their gardening efforts as well as lifestyle choices
<i>6. SP</i>	- Similar to 5th SP; tearing down the barriers to education and insufficient possibilities for personal development
<i>7. SP</i>	- Does not apply
<i>8. SP</i>	- The theoretical background and actual practical function of the Aérospot device encourages meaningful consumer choices and adds value to the individual's personal lifestyle

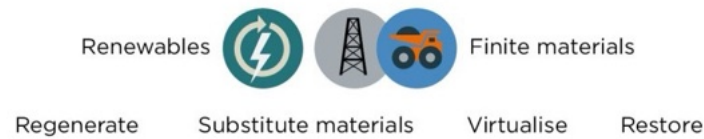
<i>Strategic</i>	- Develop a product with low energy and resource consumption and maximize output - Educate the user through data provision and intuitive guidance - Inspire consumer behaviour/lifestyle - Build a strong community with common interest
<i>Actions</i>	- Develop a prototype (from a concept to physical fully functional unit) - Build strong identity and brand awareness - Crowdfunding campaign - Keep innovating
<i>Tools</i>	- Product life-cycle analysis (LCA) - Business Canvas Model

Appendix 11 – Framework for strategic sustainable development, determining (violation of) 8 sustainability principles of Aéroport

PRINCIPLE

1

Preserve and enhance natural capital by controlling finite stocks and balancing renewable resource flows
ReSOLVE levers: regenerate, virtualise, exchange



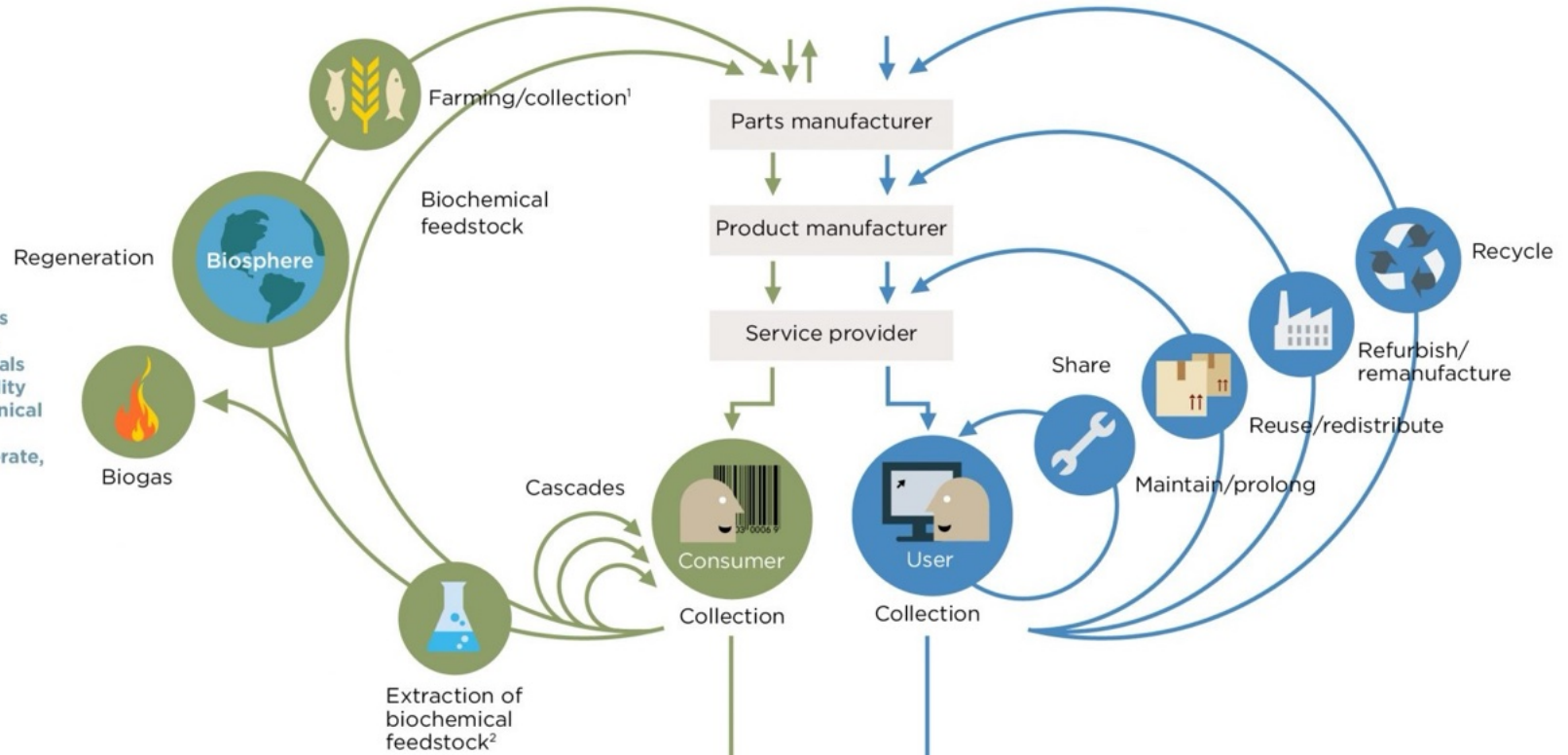
Renewables flow management

Stock management

PRINCIPLE

2

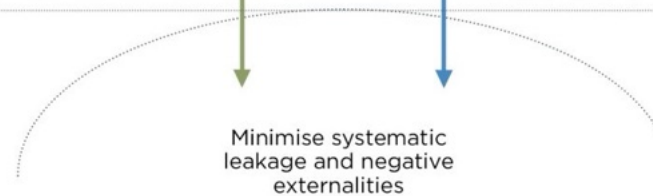
Optimise resource yields by circulating products, components and materials in use at the highest utility at all times in both technical and biological cycles
ReSOLVE levers: regenerate, share, optimise, loop



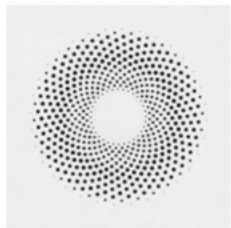
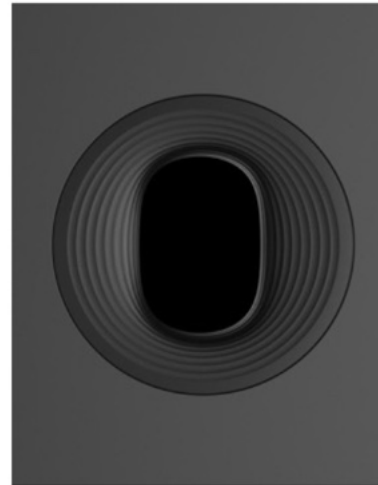
PRINCIPLE

3

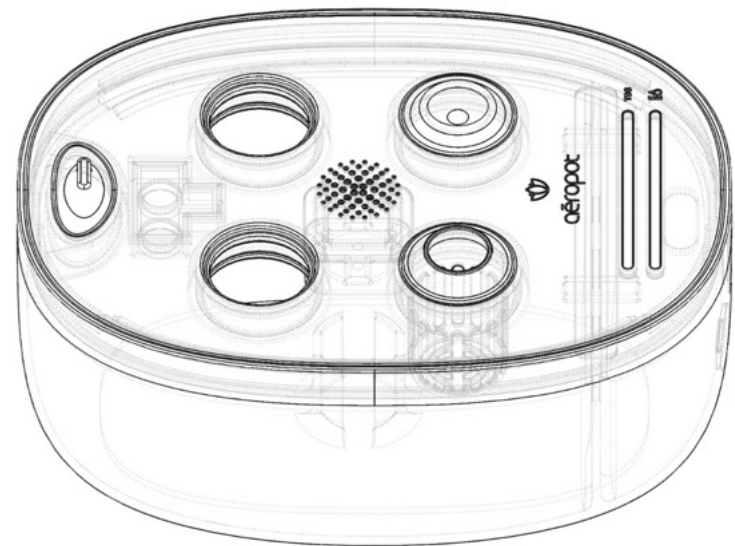
Foster system effectiveness by revealing and designing out negative externalities
All ReSOLVE levers



1. Hunting and fishing
2. Can take both post-harvest and post-consumer waste as an input



Appendix 13 – Moodboard for the outlook of the planter prototype (Pinterest, 2018)

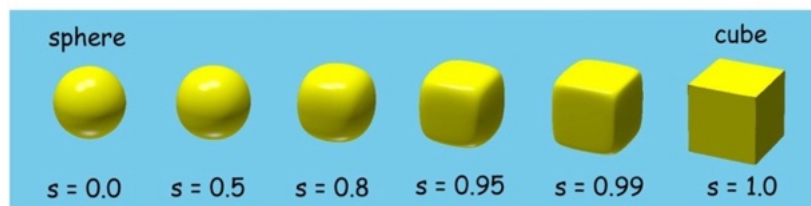
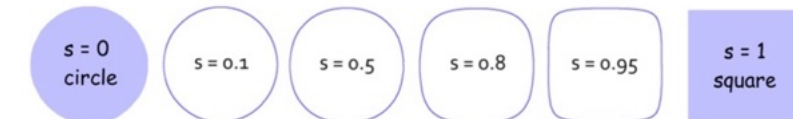
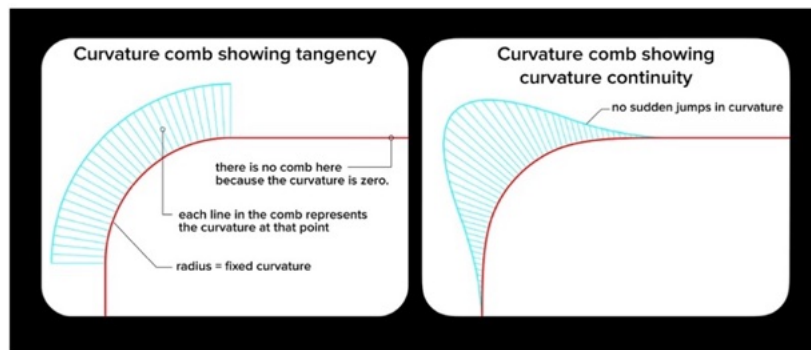
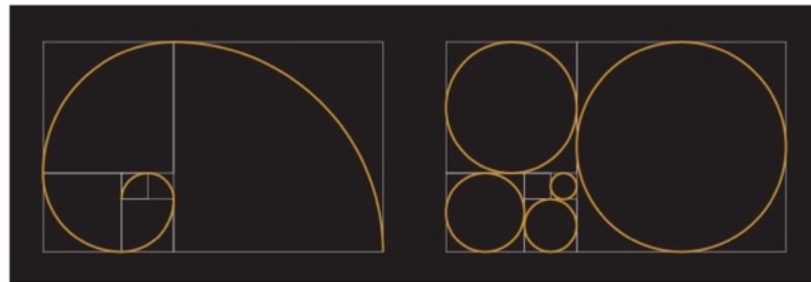


Appendix 14 – Aeropot planter prototype

Category	Subcategory	Description
<i>Function</i>	Understanding	Users know that a function exists and have no difficulty in finding it, but they don't understand how to use it.
	Finding	Users have difficulty in finding a specific function.
	Lack of need	Users don't use and need some functions. Sometimes these functions just confuse them.
<i>Performance</i>	Compatibility	Product is impossible or difficult to use with other software or hardware.
	Time	Product is annoying because it is too slow or too fast.
	Battery	Battery life is not long enough and seems to become less and less.
	Efficiency	Product is less efficient with regard to technical performance.
	Error	Sometimes an error occurs that cannot be solved through a helpdesk.
<i>Sensation</i>	Sound	Product is not loud enough to listen to or is too noisy.
	Tactility	Users feel unpleasant touching or using product.
	Weight	Product is heavy to carry or use.
	Heat	Users feel unpleasant because of heat emitted by product.
	No sense	User cannot feel any differences between options or levels.
<i>Health</i>	Fatigue	Users feel tired or fatigued in a part of their body while using product.
	Safety	Users are worried about harming their health.
<i>Mechanism</i>	Cable	Product is annoying because of its cable.

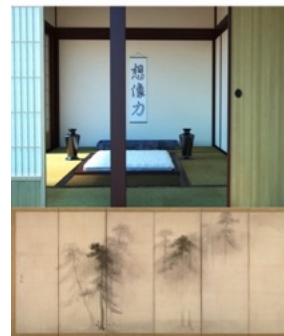
	Structure	Users feel uncomfortable because of product's mechanical structure.
	Shape	Product is too small or big to comfortably use or press. Problems occur because of its exterior form.
	Connection	Ejecting or connecting is irritating.
<i>Maintenance</i>	Service	It is difficult to get help or support. Software support is seldom updated.
	Cleaning	It is annoying to clean product.
	Care	Product requires more care.
	Durability	Product is not strong or durable enough.
	By-product	Product produces by-products that need to be managed.
<i>Constraint</i>	Lack of function	Users feel a need for a specific feature or function, with which the product would be more convenient to use.
	Lack of improvement	Product is not improved compared with its previous version.
	Insufficient information	There is no feedback or feedforward in use.
<i>Trend</i>		Product's design soon becomes boring or old-fashioned.
<i>Third party</i>		The problem comes not from the product itself but from a third party.

Appendix 15 – Overview of soft problems in consumer electronic products (Kim et al., 2007)



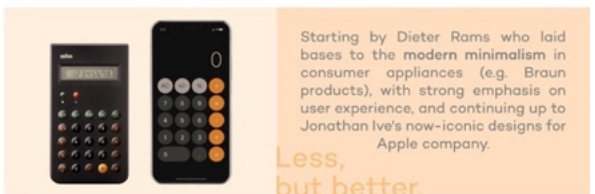
MINIMALISM JAPANESE DESIGN SCANDINAVIAN DESIGN ORGANIC DESIGN

The concept of minimalism is to strip everything down to its essential quality and achieve simplicity. The idea is not completely without ornamentation, but that all parts, details, and joinery are considered as reduced to a stage where no one can remove anything further to improve the design. The basic geometric forms, elements without decoration, simple materials and the repetitions of structures represent a sense of order and essential quality.

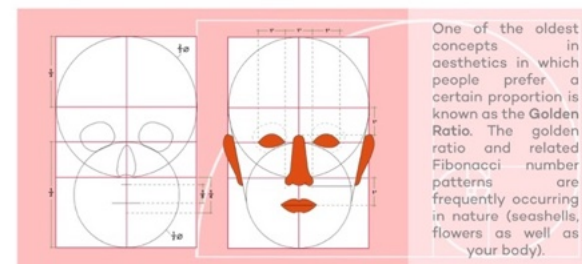


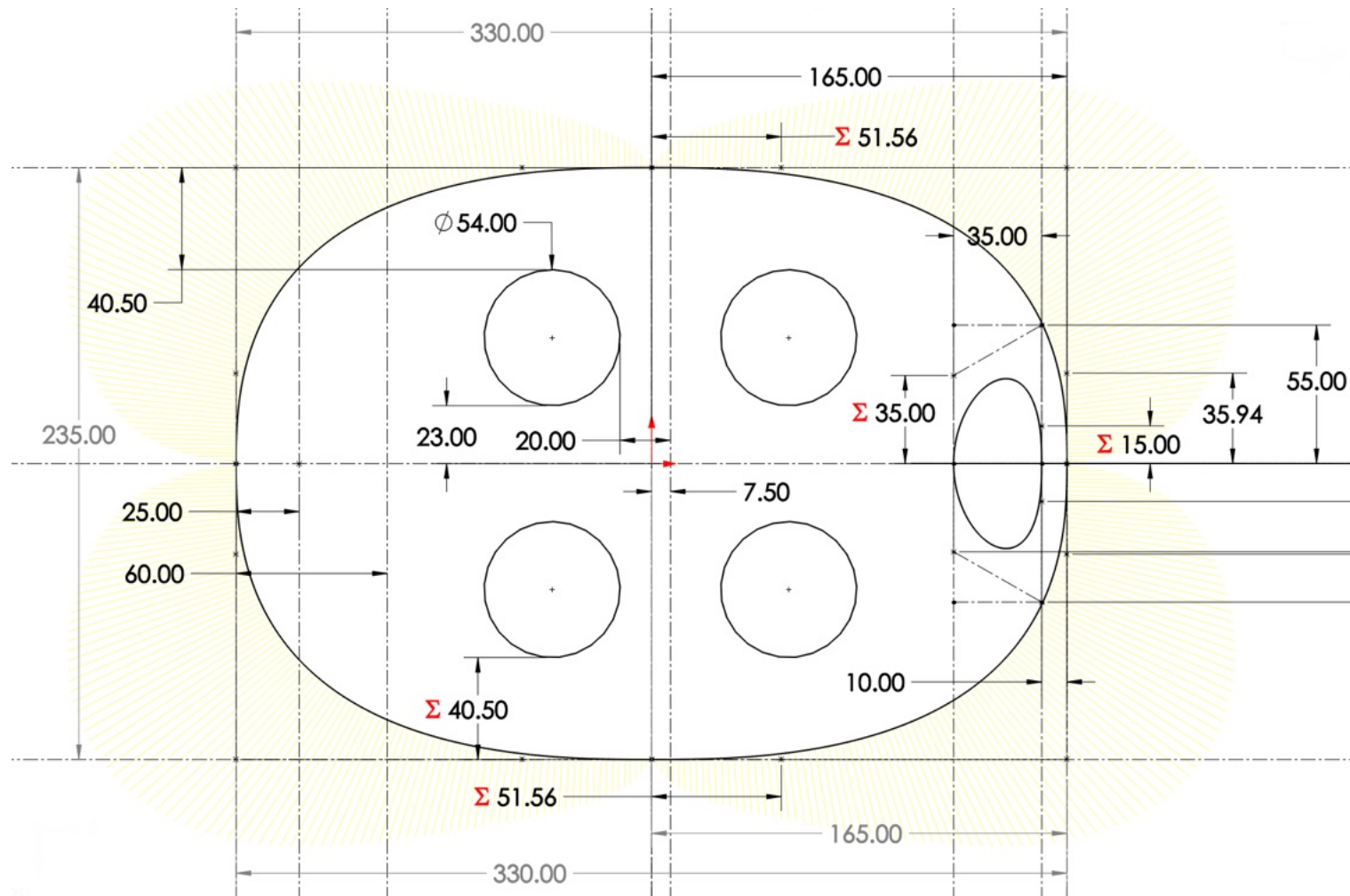
The idea of simplicity appears in many cultures, especially the Japanese traditional culture of Zen Philosophy. Japanese manipulate the Zen culture into aesthetic and design elements for their buildings. Zen concepts of simplicity transmit the ideas of freedom and essence of living. Simplicity is not only aesthetic value, it has a moral perception that looks into the nature of truth and reveals the inner qualities and essence of materials and objects.

Scandinavian design is the forerunner for democratisation of design, producing products that are beautiful, accessible, and affordable and upholding design principles that use natural materials, clean lines, and straightforward construction.



Looking for "organic" forms that are inspired and possibly aligned with the natural systems. Organic minimalism style references the curves and shapes typically found in nature. It celebrates the harmonious balance between human habitation and the organic world.



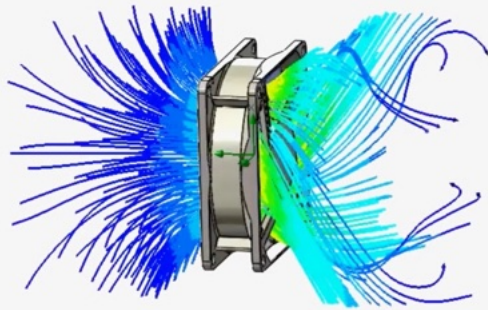


Appendix 17 – CAD sketch of the upper top part

Diffused LED lighting (top cover)



Air circulation (oxygenation)



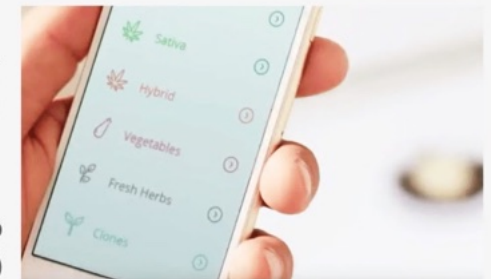
LED level indicator
(liquid, composition)



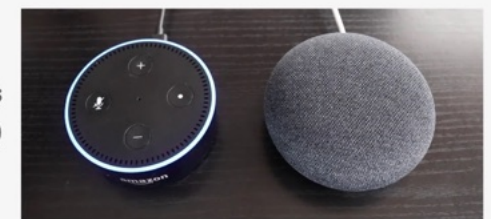
Aeroponics (ultrasonic misting)



Smartphone app
(iOS, Android)

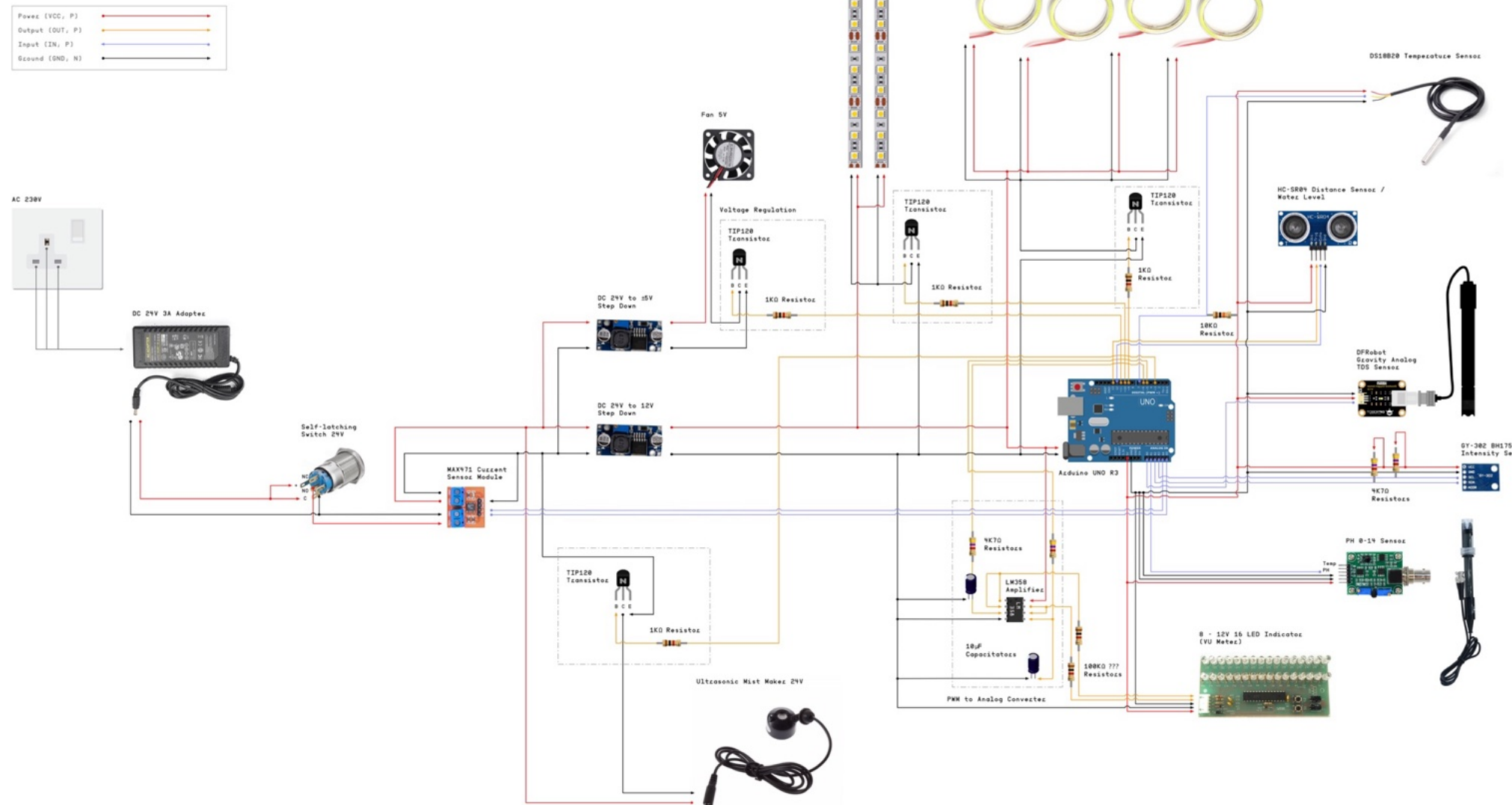


Internet of Things
(Google Home, Alexa)

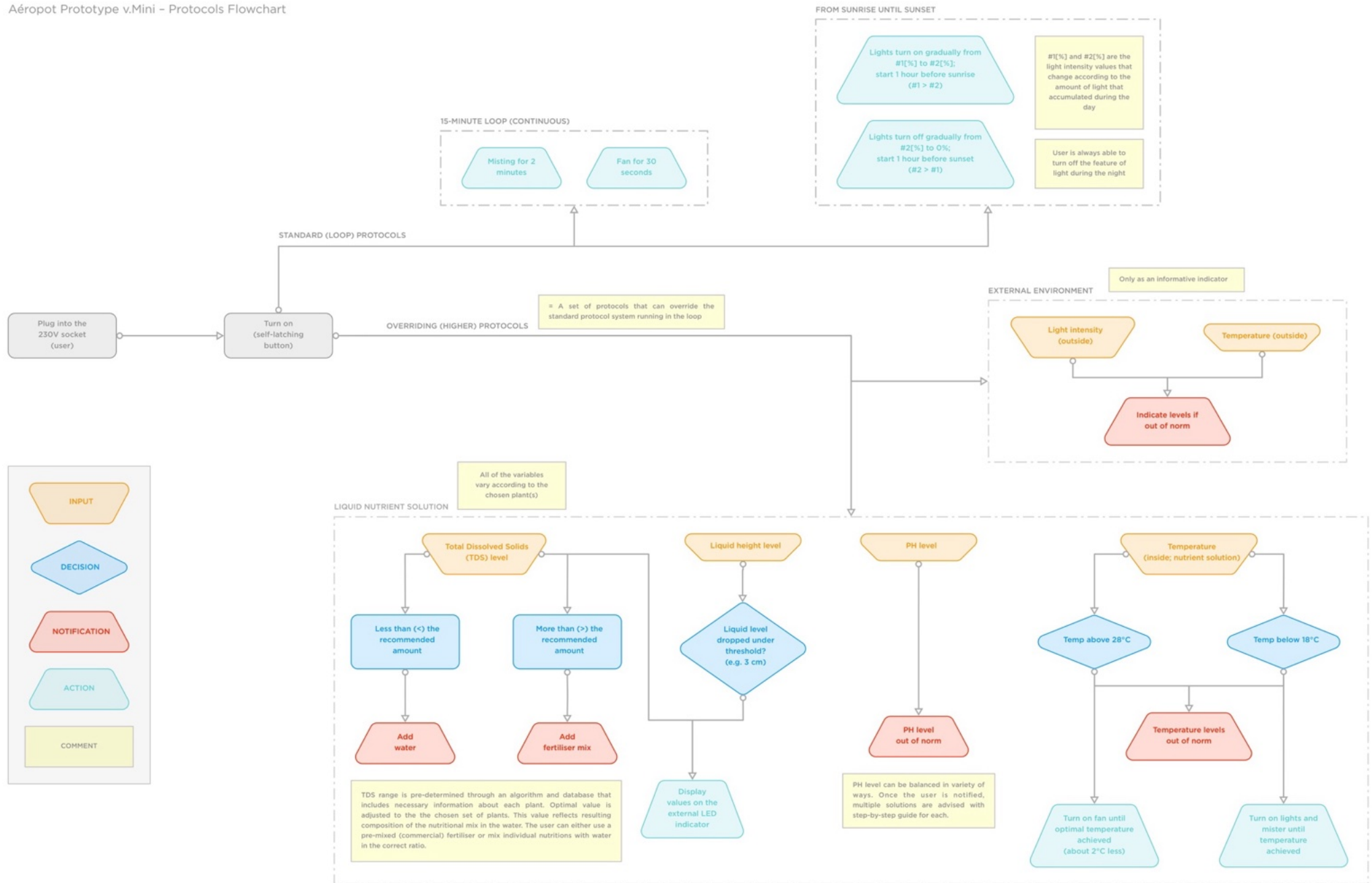


Appendix 18 – Main mechanisms/features of *Aéropot* planter

Aéropot Prototype v. Mini - Wiring Schematic 1.1



Appendix 19 – Wiring schematic of the Aéropot system





Appendix 21 – The final prototype version of the Aérospot planter

Meter/sensor	Units	MiliQ Water	Tap water (lab)	NaCl 0.01g/100ml	NaCl 0.1g/100ml	Formulex 0.1ml/100ml	Formulex 1ml/100ml	CH ₄ N ₂ O 0.1g/100ml	CH ₄ N ₂ O 1g/100ml
Temperature	°C	23.7	21.4	23.7	24.3	23.7	24.4	23.9	24
HI-98129 EC	µS/cm	1	680	192	2,085	236	1,745	3	1
DFRobot EC		0	683	261	1,850	344	1,730	0	0
Mi Flora EC		0	810	249	2,500	298	2,200	0	0
Mi Meter TDS	ppm	1	339	97	1,030	122	856	1	1
HI-98129 pH	pH	8.3	7.58	8.32	7.38	4.82	3.85	5.83	6.27
Sensor 0-14 pH		8.1	7.71	7.9	7.36	4.95	4.15	6.15	6.72
Delta 350		6-8 (!)	7.75	6-7 (!)	5.9 (!)	4.87	4	6	6.63

Meter/sensor	Units	KH ₂ PO ₄ 0.01g/100ml	KH ₂ PO ₄ 0.1g/100ml	K ₂ SO ₄ 0.01g/100ml	K ₂ SO ₄ 0.1g/100ml	KH ₂ PO ₄ 1g/1000ml	K ₂ SO ₄ 1g/1000ml	KH ₂ PO ₄ 1g/1L + K ₂ SO ₄ 1g/1L	0.192g KSO ₄ , 0.328g NH ₄ H ₂ PO ₄ , 0.349g CH ₄ N ₂ O
Temperature	°C	24	23	23.5	23.9	24.2	24.3	24	24.5
HI-98129 EC	µS/cm	107	789	172	1,491	802	1,412	1,092	545
DFRobot EC		158	960	250	1,618	1,068	1,870	1,446	730
Mi Flora EC		141	970	227	1,950	1,050	1,950	1,542	730
Mi Meter TDS	ppm	55	410	88	750	405	704	552	273
HI-98129 pH	pH	5	4.61	6.48	3.41	4.6	3.82	4.18	4.47
Sensor 0-14 pH		5.26	4.85	6.65	3.72	4.92	4.12	4.45	4.7
Delta 350		5.16	4.8	7	3.6	4.8	3.93	4.3	4.55

Appendix 22 – Data from the lab tests (grams of solute / millilitres of solvent = Mili-Q water)

Product							
	Click & Grow	AeroGarden	Plantui	123WeGrow	Ikea Växer	Akarina	U-ING
Company website	clickandgrow.com	aerogarden.com	plantui.com	123wegrow.com	ikea.com	akarina.uk	uing.u-tc.co.jp
Type	Passive hydroponic (+ “Smart soil”)	Passive hydroponic	Active hydroponic (sprayer)	Hydroponic/Aeroponic	Passive hydroponic	Passive hydroponic	Passive hydroponic
Value proposition	“Smartpots that take care of plants automatically with no external assistance needed.”	“Desktop hydroponic systems for the home consumer.”	“The fully automated indoor garden lets you grow tasty, healthy & fresh greens without soil.”	“Soil-free, Hydroponic indoor garden, that grows plants 5 times faster than soil.”	“Our indoor gardening kits, cultivators and accessories help you sprout in any space and keep your garden growing.”	“Combining contemporary Japanese product design with innovation of concept, this model adds style and sophistication to any home interior.”	“Cultivate herbs, lecture and other vegetables in your home so you always have fresh greenery for your salads and meals.”
Number of plants	Two versions: for 3 or 9 plants	Multiple versions: from 3 to 9 plants	Two versions: for 3 or 6 plants	Two versions: for 3 or 5 plants	8 plants	Wide model range: up to 16 plants	6 plants
LED light	✓	✓	✓	✓	✓	✓	✓
Automated	✓	✓	✓	✓	X	X	✓
App	✓	✓	X	X	X	X	X
Cost	€99.95/€199.95	\$50.00 - \$500.00	€149.00/€265.00	79.00/\$149.00	£47.50	£90.00-£200.00	£170.00

Product						
	Garland	Easirco	Meigar	IWS Oxypot	Atami Wilma	Platinum Aerostar
Company website	garlandproducts.com	aliexpress.com	aliexpress.com	growell.co.uk/growing-systems/hydroponics-growing-systems/iws-oxy-pot.html	atami.com	platinumhydroponics.com
Type	Passive hydroponic	Ultrasonic	Hydroponic (pump)	Hydroponic (pump)	Hydroponic (pump)	Active hydroponic (sprayer)
Value proposition	“Grow what you love, love what you grow - all year long.”	X	“Simple and easy to use, and will have you growing top quality hydroponic plants in no time.”	“You can expect explosive root growth & staggering yields.”	“The Wilma systems ensure that your plants are supplied with nutrient water in a professional manner, so that the plants will not lack anything and can perform optimally.”	“The dream of taste aeronomics at an economical price and simple solution to use.”
Number of plants	4 trays	8 plants	Multiple version: from 2 to 24 plants	Multiple versions: from 1 to 8 plants	Two versions: from 4 to 20 plants	Wide model range: from 4 to 200 pots
LED light	✓	✓	X	X	X	✓
Automated	X	✓	X	X	✓	✓
App	X	X	X	X	X	X
Cost	£40	\$50.00	\$25.00-\$75.00	£30.00-£80.00	£70.00-£200.00	€95.00-€1200.00

Appendix 23 – Competitor analysis (company websites)

Key Partners Key suppliers: 1. Electronics supplier(s) - All components are available on the market - One way is to get discounts from B2C marketplace platforms (e.g. Alibaba) or make agreements with individual suppliers (not necessarily from Asia) - e.g. Arduino, Raspberry Pie, Bigclown 2. Case manufacturer - Injection moulding of the plastic parts Collaborators: 1. Central Research Lab - Accelerator - Connection to suppliers, retailers, and investors - Increase reputation and reach of the company/company - All-round support 2. Brunel University - Help from academic experts - Provision of tools and facilities Building relationships with companies and other entities/initiatives is crucial for gaining credibility and reputation with a final goal to reach more customers.	Key Activities Design and engineering: - To achieve a high-quality product that is reliable and efficient - In order to be competitive and bring actual value to the customer - Practical usefulness and intuitive user experience have the highest priority - If the product is great, there is less need for an extensive marketing strategy Marketing: - Brand development/management - Marketing material: manual, showcase images/videos, company website, social media platforms - Collab with other companies - Showcasing events Production and logistics: - Manufacture in time with the right quantity - Delivery to a customer in a timely manner + customer support Key Resources Physical resources: 1. Electronics - Manufacturing of the case (chassis) outsourced - Construction, testing, packaging and distribution are done in-house 2. Distribution networks - Workshop/office necessary - Delivery and collection partners Other: 1. Intellectual resources - Brand, patents and copyrights 2. Team - Well-balanced with necessary skills	Value Propositions Aéropot is an aeroponic planter. It is an integrated (all-in-one) multi-functional solution with a simple design language and smart control system. It is connected to a smartphone app with an intuitive user interface. It is a tool for ecological education that is empowering individuals to get true data and understanding about their gardening efforts. Multiple functions or ways to use: - Flower caring/cultivation, adding aesthetic value - Air purification, oxygen production - Edible plants List of benefits: - Convenience, energy and resource saving - Soil-less = requires less water (90% less than soil and 70% less than classic hydroponics) and decreases the growing cycle - Low selling price point (or comparable to existing products on the market) = £80.00 to £80.00 - Highly customizable app functions/options/settings - Unobtrusive design (i.e. LED implemented in the case, magnetic opening, no cables, etc.) - Different versions of the product (e.g. minimal, essential, smart) - Strong brand identity (i.e. design language, mission, brand values, etc.) - High-tech, intuitive control, sleek design, high-quality materials and build structure - Accurately monitors the nutritional intake of the plants and notifies the user about the development of each plant - Allows for easy maintenance and customization - Modular construction (e.g. option to add extra sensors, different case colours, allows for easily swapping broken parts, etc.) - Intuitive and modern-looking smartphone app for iOS and Android - Allows for connection to 3rd party services (e.g. Google Home or Alexa through IFTTT)	Customer Relationships Online presence: - Presentation through the company website and social media - Marketing campaigns and other advertisement techniques = creating content - Building trust, transparency, meaning and authenticity - Crowdfunding campaign - Retailers and distributors - Online news/magazine articles Community: - Support through the app or website - Public forum - Public roadmap - Suggestion for improvements from customers Other: - Word of mouth and referrals - Partnerships Channels Online sales: - Webshop (direct) - Online marketplace platforms (e.g. Amazon) - Partner's customer channel User account: - Created once the product is bought - Used for the login to the smartphone and all other associations with the product (e.g. control setting and monitoring data) - Using data from sensors for service improvement and function precision (to build up best practice scenarios)	Customer Segments Gardening enthusiasts: - Interest in hydroponics/aeroponics IoT enthusiasts - Early technology adopters - Owners of smart technology gadgets - Appealing aspect of 'independence' and increased quality of life Millennials: - Technology savvy - Healthy lifestyle Older generation: - Baby boomers - Actively taking care of plants (either outdoor or indoor) Offices: - Aesthetic and well-being value Children: - Educational tool Market trends: - Germany and the United Kingdom are the largest gardening markets in Western Europe, accounting for a 37% value share - The UK has the most lucrative market, consumer spending is forecasted to grow by 22% and reach £7 billion by 2021 The market is ripe for innovative gardening solutions as the urbanisation levels keep rising and residents have limited access to outdoor gardens. 67% of the world's population will live in the cities by 2050. Already, almost 20% of city dwellers have no outside space where they live. Moreover, research suggests that companies will invest in products that allow users to save time and money, and curb the effort needed to take care of the garden. This will include a focus on Internet of Things (IoT) technology. Efficient and innovative gardening solutions, such as hydroponics and "smart gardening" will drive innovation.
Cost Structure Development - Software and hardware - Legal compliance (e.g. safety certificates) Manufacturing - Components - Labour and construction process Distribution: - Agent's fee Other: - Day to day costs - Hosting, salaries, utilities, rent	Revenue Streams Asset sale: - Webshop (B2C) - Online marketplace retailers (e.g. Amazon or eBay) - Other retailers (B2B and B2B2C) - Affiliate commissions - Accessories sale (e.g. sensors, different lights, etc.) - Sell multiple versions (differentiated by colour and/or function) Subscription: - A specifically tailored nutritional solution is sent to the customer - The composition of the fertilizer and the timing of the delivery calculated in accordance with the user's combination of plants - Opt-in option - A yearly plan Investment: - Crowdfunding campaign - Potential investors - Accelerators and competition prizes			

Appendix 24 – Business canvas model

Component	Price (per unit)	Quantity	Percentage
DFRobot Gravity Analog TDS Sensor	£15.02	×1	28.85%
LED Halo Ring 80mm 12V Red/Blue	£2.80	×4	21.51%
BNC PH 0-14 Sensor	£11.03	×1	21.18%
Power Supply Adapter 24V 3A	£7.10	×1	13.64%
LED Strip 50cm 12V Red/Blue (3:1)	£0.86	×4	6.61%
16 LED Indicator (VU Meter) 8 - 12V	£2.63	×1	5.05%
Ultrasonic Mist Maker 24V	£2.51	×1	4.82%
Arduino UNO R3	£2.45	×1	4.71%
XL4015 DC Step Down	£1.05	×2	4.03%
LED Self-locking Push Button 12V	£1.95	×1	3.74%
Cooling Fan 40mm 5V	£1.19	×1	2.29%
DS18B20 Temperature Sensor	£0.93	×1	1.79%
GY-302 BH1750 Light Intensity Sensor	£0.89	×1	1.71%
MAX471 Voltage/Current Sensor	£0.88	×1	1.69%
HC-SR04 Distance Sensor	£0.78	×1	1.50%
Total cost:	£52.07		
+ Complementary components - resistors, transistors, diodes, breadboard, cables, connectors, glands, etc.			
+ Tools - soldering station, glue gun, multimeter, etc.			

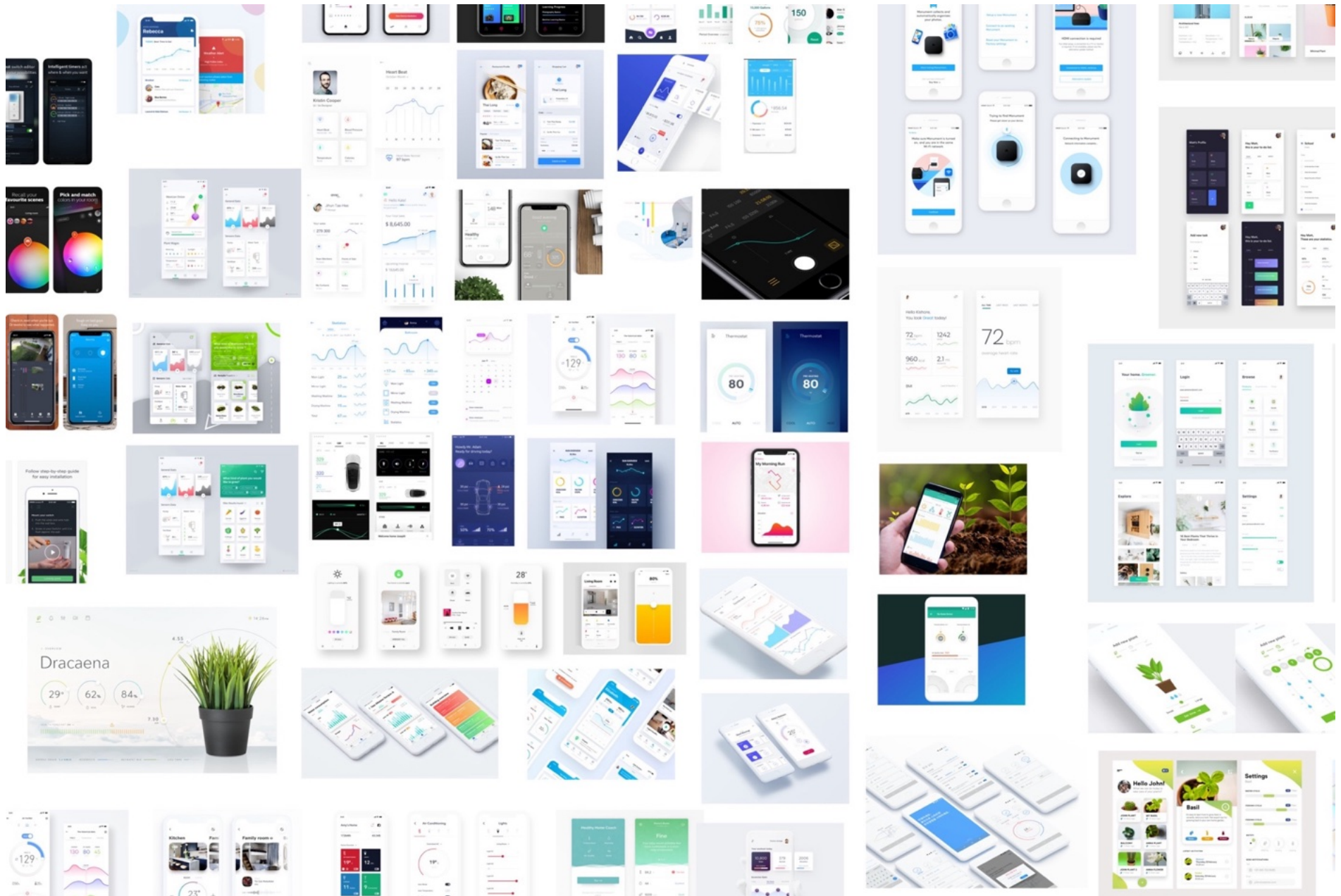
Appendix 25 – Aéroport prototype cost breakdown by component

Product	Model	Quantity	Price	Total
500g Mono ammonium phosphate fertilizer	500g	1	£4.98	£4.98
500g Urea	500g	1	£3.06	£3.06
500g Potassium Sulphate-water soluble	500g	1	£4.02	£4.02
Sub-Total:				£10.05
Second Class Standard Post (Weight: 1.50kg):				£4.50
VAT (20%):				£2.91
Total:				£17.46

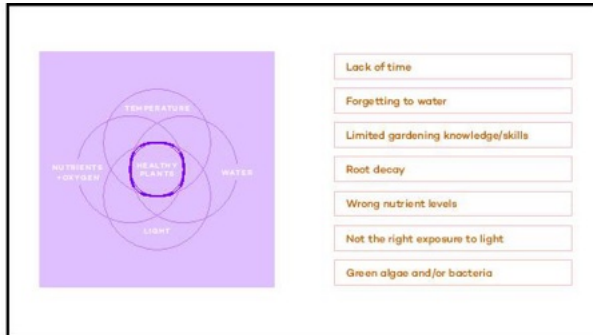
Appendix 26 – Cost of NPK compounds from Minerals Water company



Appendix 27 – 3D printed planter + fitting of electronics 2. Testing ultrasonic mister 3. Lab nutrient tests 4. Painting workshop



Appendix 28 – Moodboard for the app UI (Dribbble, 2018)



Effective technology

Unobtrusive design

Energy and resource saving

Accurate monitoring and control

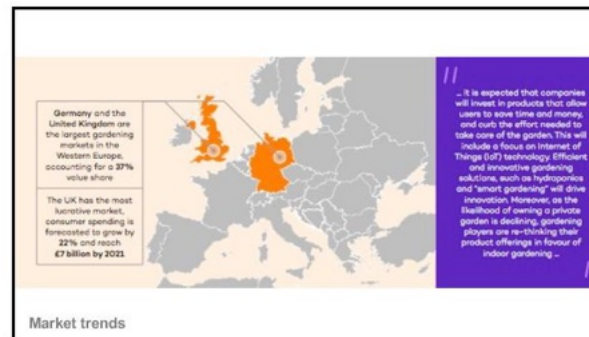
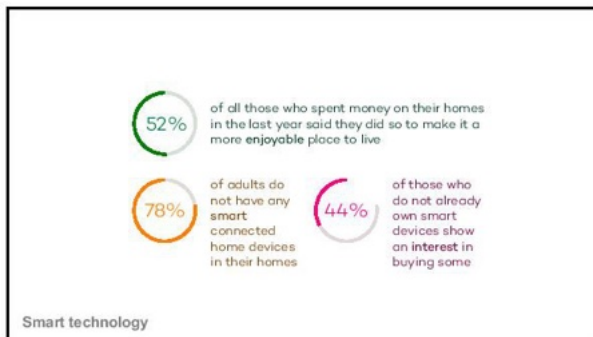
Requires less water (90% less than soil and 70% less than classic hydroponics)

Easy maintenance and customization

Decreases the growing cycle of plants

Allows for connection to 3rd party services (IoT)

Who is the customer?



Gardening enthusiasts

IoT enthusiasts

Millennials

Older generation

Offices

Children

Business model

Revenue streams

Crowdfunding campaign

Subscription

Webshop (B2C)

Accessories

Retailers (B2B2C)

Multiple versions

